High Volatility Is the Key: A Joint Treatment of Asset Pricing Puzzles*

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Abstract

Increasing evidence indicates that the marginal investor is not the average consumer, whose consumption volatility is low. This article shows that in a single model with just one state variable and a higher level of consumption volatility, five central asset pricing puzzles disappear: the equity premium, risk-free rate, real bond premium, excess volatility, and return predictability. The calibration takes the marginal investor to be a financial intermediary and measures its consumption volatility from dealer returns. With only two free parameters, the model matches all eight targeted moments.

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The asset pricing literature has accumulated a long list of puzzles. This article suggests that most of them disappear under a simple and natural change to one important assumption. In particular, the puzzles arise when the marginal investor is identified with the average consumer. When the marginal investor is not the average consumer, reasonable specifications exist under which they do not arise. The model in this article is mostly a standard consumption-based model, and its essential feature is the high and time-varying consumption volatility of the marginal investor. The model has a single state variable that drives both consumption volatility and consumption drift at the same time. Several parameters are set directly from relevant data or on theoretical grounds. The volatility schedule itself is measured from the equity returns of New York Fed primary dealers (He, Kelly and Manela 2017). Only two parameters are then chosen to match asset prices, against eight moments. What comes out is a model in which the equity premium, the risk-free rate, the real bond premium, excess volatility, and return predictability no longer present puzzles, and in which each of the eight targeted moments lies within one and a half model-implied standard errors of its counterpart in the data.

Several independent strands of literature indicate that there are agents other than ordinary households pricing risky assets. For instance, regular households have limited stock market participation (Haliassos and Bertaut 1995; Campbell 2006). Wealthy households, who are more plausibly marginal in asset markets, have consumption that is both far more volatile than the aggregate and more strongly correlated with stock returns (Ait-Sahalia, Parker and Yogo 2004). Supply-and-demand-driven approaches to asset pricing show that agents such as "arbitrageurs" play a special role in price formation (Vayanos and Vila 2021). The intermediary asset pricing approach identifies financial intermediaries as the relevant marginal investors (He and Krishnamurthy 2013).

The goal of this article is to address five central asset pricing puzzles, of which Gabaix (2012) gives a good summary. The equity premium puzzle, introduced by Mehra and Prescott (1985), is the finding that excess stock market returns cannot be explained by the standard consumption-based model with moderate risk aversion when calibrated to aggregate consumption data. Attempting to resolve this puzzle by raising risk aversion implies a risk-free rate far above the one observed, a further inconsistency known as the risk-free rate puzzle (Weil 1989). Furthermore, stock returns demonstrate high variability and predictability (Shiller 1981; Campbell and Shiller 1988), giving rise respectively to the excess volatility

puzzle and the return predictability puzzle.

Apart from stocks, in the data, long-term bonds also exhibit positive excess returns, which are large and time-varying, while most standard theoretical models predict excess returns for bonds that are near zero and negative. This is the bond premium puzzle (Backus, Gregory and Zin 1989).[1] These bond features are also closely related to the term structure of interest rates, as positive bond premia are associated with an upward sloping yield curve. Originally, the puzzle was introduced for nominal bonds. For real (inflation-adjusted) bonds, evidence is mixed. The real premium appears negative in the UK (Evans 1998; Piazzesi and Schneider 2006; Joyce, Kaminska and Lildholdt 2012), while in the USA most evidence points to a significant, time-varying, and mostly positive real term premium (Haubrich, Pennacchi and Ritchken 2012; Abrahams, Adrian, Crump, Moench and Yu 2016; d'Amico, Kim and Wei 2018; Kim, Walsh and Wei 2019; Pflueger and Viceira 2016; Andreasen, Christensen and Riddell 2021). Hence, generating such a premium is a key objective of this article.

Notably, addressing the bond premium puzzle for real bonds is particularly challenging. Real bond prices move inversely with interest rates, so if rates are procyclical, real bonds hedge consumption risk and should command a negative risk premium instead of a positive one. Even recursive utility models, despite their success in explaining the equity premium puzzle (Bansal and Yaron 2004), in general produce a negative real bond premium (Bansal and Shaliastovich 2013; Ellison and Tischbirek 2021). More generally, the forces that generate the high equity premium can make the bond premium larger in absolute value, but its sign depends on whether bond prices fall in the states where marginal utility is high. This article also argues that most of the mechanisms in the literature that deliver a positive real term premium incorporate to some degree the mechanism analyzed here (Section IV).

The standard model, calibrated to aggregate consumption, produces an equity premium many times smaller than the one in the data, a risk-free rate far above the observed one once risk aversion is raised to repair the premium, stock returns that are too smooth and too unpredictable, and a real term premium that is tiny and has the wrong sign. In the model of this article, these mismatches do not appear, and there is also positive evidence from the model's conditional predictions, since the model-implied prices at the state recovered

[1] A bond premium here is the short-horizon version of the term premium, the instantaneous expected excess return of Equation (12), while the term premium of Equation (13) is the premium over a holding horizon equal to the bond's remaining maturity. In this model one implies the other, so the two terms are used interchangeably except where the distinction matters.

from dealer volatility alone comove positively with observed price–dividend ratios and the yield spreads of Treasury Inflation-Protected Securities (TIPS). The main shortfall is the Sharpe ratio of equities, which is about half its observed value, because the model reaches the premium partly through return volatility above that of the data (Section III.G).

The model uses an exogenous process for the consumption of the marginal investor, so it offers no endogenous account of why that consumption is so volatile, and the frictions that would generate it are left outside. The aim of the article is instead to isolate the mechanism that underlies many of the models that do generate such volatility. The level of that volatility comes from the data, since the schedule is measured from dealer equity returns, and it is what separates the marginal investor from the representative household. The calibrated structure implies a specific interplay between the consumption drift and the diffusion, which is what delivers the results of the article. In particular, in connection with the real term premium, the real rate rises with consumption volatility rather than moving procyclically, and that is what makes long bonds risky. A negative consumption shock raises the volatility state, which raises the real rate and depresses bond prices, so bonds lose value exactly when the marginal investor's consumption has fallen.

The calibration, described in Section II, measures the volatility schedule from the dealer equity returns, whose explanatory power for expected returns across many asset classes is documented by He *et al.* (2017). Dealer equity volatility is treated as the return volatility of the marginal investor's wealth in the model, and the model-consistent consumption volatility schedule is derived from it. Notably, the factor of He *et al.* (2017) is the shock to the dealers' equity capital ratio, and where they let the price of risk vary over time they model it as a function of the level of that ratio. The volatility of dealer equity returns, measured from the returns themselves, is a new measure here, and its relevance is shown for both the conditional and the unconditional predictions of the model. More broadly, Adrian, Etula and Muir (2014) and Haddad and Muir (2021) show that intermediary conditions drive risk premia. Whether intermediary leverage is procyclical or countercyclical is itself contested, and the two answers imply opposite signs for the correlation between the marginal investor's consumption and the volatility state. This article shows that either sign can deliver a positive real term premium, but that the one implied by countercyclical leverage fits the moments considerably better. He *et al.* (2017) document countercyclical leverage at the holding-company level, that of the publicly traded parents of the primary dealers, where the data used here are measured.

Because the theory is silent on who the marginal investor is, several interpretations are admissible. It could be a financial intermediary whose risk exposure rises as its performance deteriorates in high-volatility states. It could be a group of wealthy households whose consumption is volatile and strongly linked to financial markets (Ait-Sahalia *et al.* 2004). It could also reflect households that cannot share idiosyncratic risk, as in Mitra and Xu (2024), where limited risk sharing produces positive real bond premia. The theory does not depend on which interpretation holds. The intermediary interpretation is adopted at the level of the calibration, and the volatility measured from intermediary equity returns is enough for the model to match the moments.

The article is motivated by the evidence that the average consumer is not the marginal investor, but it does not settle that question. Its claim is that there are marginal investors whose consumption volatility is high, and that their presence is enough to account for the puzzles. Regular households may play no role in pricing assets, in which case what remains to be explained is why they neither participate in those markets nor otherwise influence them. Alternatively, in heterogeneous-agent models, regular households could be subject to habit formation (Campbell and Cochrane 1999), long-run risks (Bansal and Yaron 2004), rare disasters (Barro 2006; Gabaix 2012), or other mechanisms of the representative-agent literature, while high-volatility investors are also present. These mechanisms would then complement the one studied here rather than compete with it, and the crucial question would be how and to what extent the two types of agents share risk.

Section I sets out the model, and Section II describes the calibration, including the measurement of consumption volatility from dealer returns. Section III presents the results for each puzzle, the evidence from the state recovered from dealer volatility, and the model's limits. Section IV relates the mechanism to others in the literature, and Section V concludes.

I. Model Setup

The model focuses on a single agent who is the marginal investor, abstracts from non-marginal investors, and takes the consumption process of this agent to be exogenous. Theoretically, this could be the only agent in the economy, but it is more realistic to think that other agents also exist and, for reasons outside the model, do not participate in or affect the same asset markets. Treating consumption as exogenous follows the practice of the consumption-based

literature (Campbell and Cochrane 1999; Bansal and Yaron 2004; Wachter 2006), which asks what consumption process is consistent with asset prices. Here, rather than the average consumer, that question concerns the marginal investor, whose identity is left open until the calibration, which adopts the intermediary interpretation.

A. Utility

The lifetime utility of the agent is given by:

$$U_0 = \mathrm{E}_0 \int_0^\infty e^{-\rho t} \frac{C_t^{1-\gamma} - 1}{1-\gamma} \mathrm{d}t, \tag{1}$$

where C_t is the consumption flow at time t, γ is the coefficient of relative risk aversion, and ρ is the time preference rate. In the model, time is continuous, utility is time-separable, and the utility flow exhibits constant relative risk aversion (CRRA).

B. Consumption and State Processes

The consumption flow C_t is exogenous and follows a stochastic process:

$$\mathrm{d}\log(C_t) = \mathrm{d}c_t = \mu_c(x_t)\mathrm{d}t + \sigma_c(x_t)\mathrm{d}W_{ct} \tag{2}$$

where c_t is log consumption, x_t is the state variable, W_{ct} is a Wiener process associated with consumption, $\mu_c(x_t)$ is the consumption drift, and $\sigma_c(x_t)$ is the consumption diffusion, abbreviated below as μ_{ct} and σ_{ct}. The specific functional forms of μ_{ct} and σ_{ct} are given in Appendix B. They depend on parameters whose calibration is described in Section II. In the calibrated model both are increasing in the state variable x_t.

There is only one state variable x_t, and it follows a mean-reverting process:

$$\mathrm{d}x_t = \log(\phi)x_t\mathrm{d}t + \sigma_x\mathrm{d}W_{xt} \tag{3}$$

$$\mathrm{d}W_{xt}\mathrm{d}W_{ct} = \rho_{cx}\mathrm{d}t \tag{4}$$

where σ_x is the state variable diffusion, W_{xt} is a Wiener process associated with the state variable, and ρ_{cx} is the correlation between the noise processes for consumption and the

state variable. The drift term shows that the steady state is at $x_t = 0$, and the process always reverts to the steady state given that $0 < \phi < 1$, as in an AR(1) with autoregressive coefficient ϕ.

C. Stochastic Discount Factor

Based on the utility function, the stochastic discount factor (SDF) is given by:

$$\Lambda_t = e^{-\rho t} C_t^{-\gamma} \tag{5}$$

Then, based on the consumption process and by applying Itô's lemma, the dynamics of the SDF can be derived, following the notation and the approach of Cochrane (2009):

$$\frac{\mathrm{d}\Lambda_t}{\Lambda_t} = \Big(-\rho - \gamma\mu_{ct} + \frac{\gamma^2}{2}\sigma_{ct}^2 \Big)\mathrm{d}t - \gamma\sigma_{ct}\mathrm{d}W_{ct} \tag{6}$$

The instantaneous risk-free rate then follows as:

$$r(x_t)\mathrm{d}t = -\mathrm{E}_t\left[\frac{\mathrm{d}\Lambda_t}{\Lambda_t}\right] = \Big(\rho + \gamma\mu_{ct} - \frac{\gamma^2}{2}\sigma_{ct}^2\Big)\mathrm{d}t \tag{7}$$

D. Zero-Coupon Bond Pricing

The SDF derived above can be used to price any asset via the standard pricing equation. In particular, a zero-coupon bond that pays one unit of consumption at maturity $T = t + m$ must satisfy:

$$\mathrm{E}\big[\mathrm{d}(\Lambda_t B_t(x, m))\big] = 0, \quad B(x, 0) = 1 \tag{8}$$

where $B(x, m)$ is the price of a zero-coupon bond with a remaining maturity of m at time t. The price of the bond is a function of the state variable x_t and maturity m. Using Itô's lemma on the bond price yields the following:

$$\mathrm{d}B_t = \left(\log(\phi)x_t B_x - B_m + \frac{\sigma_x^2}{2}B_{xx}\right)\mathrm{d}t + \sigma_x B_x \mathrm{d}W_{xt}, \quad B(x, 0) = 1 \tag{9}$$

where subscripts on B denote partial derivatives with respect to the corresponding variable.[2] Using the process for the bond price, the pricing equation gives rise to a partial differential equation (PDE) (time subscripts and the arguments of B are omitted for brevity):

$$\mathrm{E}\big[\mathrm{d}(\Lambda B)\big] = 0 \Rightarrow \mathrm{E}\left[\frac{\mathrm{d}\Lambda}{\Lambda}B + \mathrm{d}B + \frac{\mathrm{d}\Lambda}{\Lambda}\mathrm{d}B\right] = 0 \tag{10}$$

$$\Rightarrow -B_m - r(x)B + \big(\log(\phi)x - \rho_{cx}\gamma\sigma_c\sigma_x\big)B_x + \frac{\sigma_x^2}{2}B_{xx} = 0 \tag{11}$$

Since $\mathrm{E}[\mathrm{d}\Lambda/\Lambda] = -r\,\mathrm{d}t$, the pricing equation gives the instantaneous expected excess return of the bond before maturity:

$$\mathrm{E}\left[\frac{\mathrm{d}B}{B}\right] - r\,\mathrm{d}t = -\mathrm{E}\left[\frac{\mathrm{d}\Lambda}{\Lambda}\frac{\mathrm{d}B}{B}\right] = \rho_{cx}\gamma\sigma_c\sigma_x\,\frac{B_x}{B}\,\mathrm{d}t \tag{12}$$

The term premium is defined as:[3]

$$TP(x_t, m) = -\frac{1}{m}\log\left[\frac{B(x_t, m)}{\mathrm{E}_t\left[\exp\left\{-\int_0^m r(x_{t+s})\mathrm{d}s\right\}\right]}\right] \tag{13}$$

Intuitively, this expresses the expected annualized excess return of the bond over a period equal to its remaining maturity. The risk-neutral yield of the bond is correspondingly defined as the yield that would prevail in the absence of any risk premium:

$$RNY(x_t, m) = -\frac{1}{m}\log\left[\mathrm{E}_t\left[\exp\left\{-\int_0^m r(x_{t+s})\mathrm{d}s\right\}\right]\right] \tag{14}$$

so that the total yield $y(x_t, m) = -\frac{1}{m}\log[B(x_t, m)]$ decomposes as $y(x_t, m) = RNY(x_t, m) + TP(x_t, m)$. Equation (12) gives the bond's expected excess return directly. Its right-hand side depends on risk aversion and on the covariance between the instantaneous changes in consumption and the bond price. In the PDE (11), $\rho_{cx}\gamma\sigma_c\sigma_x$ and $\log(\phi)x_t$ enter additively in the coefficient of B_x. The former is the term premium component, and the latter is the

[2] For simplicity the dependence on time is not explicitly shown, but maturity is connected to time through $\partial/\partial t = -\partial/\partial m$.

[3] Based on the Feynman-Kac formula, the equation can also be written as: $TP(x_t, m) = -\frac{1}{m}\log\left[\frac{\mathrm{E}_t^*\left[\exp\left\{-\int_0^m r(x_{t+s})\mathrm{d}s\right\}\right]}{\mathrm{E}_t\left[\exp\left\{-\int_0^m r(x_{t+s})\mathrm{d}s\right\}\right]}\right]$, where the numerator uses the risk-neutral measure.

expectation component, which captures expected future changes in the short rate. Because the two are additive, their relative importance for the shape of the yield curve can be compared directly, for instance to tell whether an upward-sloping yield curve reflects expected increases in the short rate or a term premium. If the term premium component is negligible compared to the expectation component, the term premium is zero for practical purposes, and if $\rho_{cx} = 0$, it is exactly zero and the expectations hypothesis holds in its strongest form.[4] The formula also shows what a sizeable positive term premium requires. The product $\rho_{cx}\gamma\sigma_c\sigma_x$ must be large in absolute value, which needs either high consumption volatility or high risk aversion, and ρ_{cx} and B_x must have the same sign. With $\rho_{cx} < 0$, as in the calibration, the bond price must fall as the state rises, which needs a real rate that increases with volatility.

E. Equity Pricing

Having defined bond prices, a similar approach is applied to stocks. First, stocks are assumed to pay a dividend that follows the process:

$$\frac{\mathrm{d}D_t}{D_t} = \mu_D \mathrm{d}t + \sigma_D \mathrm{d}W_{Dt} \tag{15}$$

$$\mathrm{d}W_{Dt}\mathrm{d}W_{ct} = \rho_{cD}\mathrm{d}t, \quad \mathrm{d}W_{Dt}\mathrm{d}W_{xt} = \rho_{xD}\mathrm{d}t \tag{16}$$

where D_t is the dividend flow at time t, μ_D is the drift of the dividend process, σ_D is the dividend diffusion, W_{Dt} is a Wiener process associated with the dividend process, ρ_{cD} is the correlation between the noise processes for consumption and the dividend, and ρ_{xD} is the correlation between the noise processes for the state variable and the dividend. For simplicity, μ_D and σ_D are assumed to be constant. The stock price is built by first pricing individual dividend strips and then aggregating across all maturities. A dividend strip with maturity m is a security that pays the dividend D_{t+m} at time $t + m$. Let $\hat{S}(x_t, m)$ denote its price at time t.[5]

It is convenient to work with the strip price–dividend ratio $\hat{s}(x_t, m) = \hat{S}(x_t, m)/D_t$,

[4] In the literature, the normal form of the expectations hypothesis refers to whether the term premium varies over time, not whether it is exactly zero.

[5] This slightly abuses notation. To be fully precise, an extra time subscript is required, that is, $\hat{S}_t(x_t, m)$, in order to specify the timing of the dividend, and the same applies to the ratio $\hat{s}$ below. In this notation, maturity occurs at time $t + m$. For brevity, maturity is taken as given and the time subscript is omitted.

normalized by the current dividend. This normalization yields a clean terminal condition. At maturity $m = 0$, the strip pays the dividend of that date, so $\hat{s}(x, 0) = 1$, which uniquely pins down the solution.[6] By Itô's lemma, $\hat{s}$ follows the process:

$$\mathrm{d}\hat{s}_t = \left(\log(\phi) x_t \hat{s}_x - \hat{s}_m + \frac{\sigma_x^2}{2}\hat{s}_{xx}\right)\mathrm{d}t + \sigma_x \hat{s}_x \mathrm{d}W_{xt} \tag{17}$$

The strip price–dividend ratio satisfies a pricing equation analogous to the bond case (time subscripts and the arguments of $\hat{s}$ are omitted for brevity):

$$\mathrm{E}\big[\mathrm{d}(\Lambda \hat{S})\big] = 0 \tag{18}$$

$$\Rightarrow \mathrm{E}\big[\mathrm{d}(\Lambda \underbrace{(\hat{s}D)}_{\substack{\text{price of}\\ \text{dividend strip}}})\big] = 0 \tag{19}$$

$$\Rightarrow \mathrm{E}\left[\frac{\mathrm{d}\Lambda}{\Lambda}\hat{s} + \mathrm{d}\hat{s} + \frac{\mathrm{d}D}{D}\hat{s} + \frac{\mathrm{d}\Lambda}{\Lambda}\mathrm{d}\hat{s} + \frac{\mathrm{d}\Lambda}{\Lambda}\frac{\mathrm{d}D}{D}\hat{s} + \mathrm{d}\hat{s}\frac{\mathrm{d}D}{D}\right] = 0 \tag{20}$$

$$\Rightarrow -\hat{s}_m - \underbrace{\left(r(x) - \mu_D + \rho_{cD}\gamma\sigma_c\sigma_D\right)}_{\hat{r}(x)}\hat{s} + \left(\log(\phi)x - \rho_{cx}\gamma\sigma_c\sigma_x + \rho_{xD}\sigma_x\sigma_D\right)\hat{s}_x + \frac{\sigma_x^2}{2}\hat{s}_{xx} = 0, \quad \hat{s}(x, 0) = 1 \tag{21}$$

where D is the current dividend D_t. Comparing Equation (21) with Equation (11) shows an analogous structure, while the differences stem from the stochastic dividend process, which adjusts the discount rate and adds a term to the drift. The risk premium of the dividend strip, used here equivalently with the expected excess return, can also be derived from the pricing equation:

$$\mathrm{E}\left[\frac{\mathrm{d}(\hat{s}D)}{\hat{s}D}\right] - r\,\mathrm{d}t = -\mathrm{E}\left[\frac{\mathrm{d}\Lambda}{\Lambda}\left(\frac{\mathrm{d}\hat{s}}{\hat{s}} + \frac{\mathrm{d}D}{D}\right)\right] = \left(\rho_{cx}\gamma\sigma_c\sigma_x\frac{\hat{s}_x}{\hat{s}} + \rho_{cD}\gamma\sigma_c\sigma_D\right)\mathrm{d}t \tag{22}$$

As with the bond, the risk premium is connected to the level of risk aversion and the covariance of the instantaneous changes in consumption with the two sources of the strip's return. The first term on the right-hand side reflects changes in the price–dividend ratio and the second

[6] Determining the price–dividend ratio of the stock directly gives a differential equation in the state variable alone. It does not require a terminal condition in time, but as a second-order equation it requires two boundary conditions (Chen, Cosimano and Himonas, 2010). The approach followed here avoids this.

reflects changes in the dividend itself. In the calibration below, the dividend correlations are set to zero, so the second term vanishes and the entire equity premium comes from valuations falling as volatility rises, with no contribution from cash-flow risk. The stock price–dividend ratio is obtained by integrating the strip price–dividend ratio across all maturities, as in Wachter (2006):

$$\frac{S_t}{D_t} = s(x_t) = \int_0^\infty \hat{s}(x_t, m)\mathrm{d}m \tag{23}$$

where the stock's dividend flow matches the dividends paid by the dividend strips.[7] The expected excess return of the stock is the weighted average of the strip excess returns across maturities, which by Equation (22) takes the same form as for the bond and the strip:[8]

$$\begin{aligned} \mathrm{E}\left[\frac{\mathrm{d}S}{S}\right] + \frac{D}{S}\,\mathrm{d}t - r\,\mathrm{d}t &= \frac{1}{s}\int_0^\infty \left(\mathrm{E}\left[\frac{\mathrm{d}(\hat{s}D)}{\hat{s}D}\right] - r\,\mathrm{d}t\right)\hat{s}\,\mathrm{d}m \\ &= \left(\rho_{cx}\gamma\sigma_c\sigma_x\,\frac{s_x}{s} + \rho_{cD}\gamma\sigma_c\sigma_D\right)\mathrm{d}t \end{aligned} \tag{24}$$

The bond and strip PDEs are solved numerically by finite differences, with zero-flux Neumann conditions at boundaries placed well beyond the states the model visits. As a check, bond prices are also computed by Monte Carlo simulation of the state process, and the two methods agree.

F. The Link between the Consumption and the Wealth Process

The consumption process can be linked to the wealth process, since it could be the result of a wealth-management problem. The derivations in this subsection are used later in the calibration, which measures the volatility process from the equity returns of NY Fed primary

[7] In the case of a dividend strip, a lump sum amount is paid at a specific point in time, while the stock pays a dividend flow.

[8] Alternatively it can also be found directly through the pricing equation of the stock. Following Cochrane (2009), the pricing equation for a security with a dividend flow is given by $\mathrm{E}\big[\mathrm{d}(\Lambda S)\big] + \Lambda D\mathrm{d}t = 0$, where S is the price of the security, and D is the dividend flow. This equation can be rearranged to give the expected excess return.

dealers. The net worth of the investor follows a stochastic process:

$$\mathrm{d}N_t = N_t \left(\mu_N(x_t)\,\mathrm{d}t + \sigma_N(x_t)\,\mathrm{d}W_{Nt}\right) - C_t\,\mathrm{d}t \tag{25}$$

$$\mathrm{d}W_{Nt}\,\mathrm{d}W_{xt} = \rho_{Nx}\,\mathrm{d}t \tag{26}$$

where μ_N is the drift of net worth before consumption and σ_N its diffusion, ρ_{Nx} is the correlation between wealth shocks and state shocks, and the state variable x_t summarizes the investment opportunity set. For a levered financial intermediary with assets A_t and liabilities L_t, net worth is the equity of the institution, $N_t = A_t - L_t$, and $\sigma_N = (A_t/N_t)\,\sigma_A$, so net-worth volatility is directly observable as the return volatility of the institution's publicly traded equity. Because CRRA preferences are homothetic and the opportunity set depends only on x_t, the optimal consumption rule is linear in wealth, $C_t = \kappa(x_t)\,N_t$ (Merton 1969, 1971).[9] Applying Itô's lemma to $\log C_t = \log\kappa(x_t) + \log N_t$ then delivers log consumption dynamics of exactly the form (2):

$$\mathrm{d}\log C_t = \mu_c^N(x_t)\,\mathrm{d}t + \underbrace{\sigma_N(x_t)\,\mathrm{d}W_{Nt} + \frac{\kappa'(x_t)}{\kappa(x_t)}\,\sigma_x\,\mathrm{d}W_{xt}}_{=\sigma_{ct}\mathrm{d}W_{ct}} \tag{27}$$

with a drift $\mu_c^N(x_t)$ that depends on the state and a single composite diffusion term.[10] Matching the diffusion of (27) with that of (2) delivers the translation between the two processes:

$$\sigma_{ct}^2 = \sigma_N(x_t)^2 + 2\,\frac{\kappa'(x_t)}{\kappa(x_t)}\,\rho_{Nx}\,\sigma_N(x_t)\,\sigma_x + \left(\frac{\kappa'(x_t)}{\kappa(x_t)}\right)^2\sigma_x^2 \tag{28}$$

$$\rho_{cx}\,\sigma_{ct} = \rho_{Nx}\,\sigma_N(x_t) + \frac{\kappa'(x_t)}{\kappa(x_t)}\,\sigma_x \tag{29}$$

Under log utility the consumption–wealth ratio is constant, the terms in κ'/κ vanish, and the translation is exact state by state, $\sigma_c = \sigma_N$ and $\rho_{cx} = \rho_{Nx}$. Away from log utility, the κ'/κ terms reflect intertemporal hedging through the consumption–wealth ratio, and they can be

[9] The value function has the form $V(N,x) = f(x)\,N^{1-\gamma}/(1-\gamma)$, so $\kappa = f^{-1/\gamma}$ depends only on the state. Under log utility, $\kappa = \rho$, a constant.

[10] Matching the drift terms determines the wealth drift μ_N, since the consumption drift is pinned by the Euler equation.

computed from the model's solution.

The consumption–wealth ratio κ that enters the translation (28)–(29) is determined inside the model. In equilibrium, wealth equals the value of the claim to the investor's consumption stream, and this claim is priced with the same strip machinery as the stock. A consumption strip pays the consumption C_{t+m} at horizon m, so it is a dividend strip with consumption in the place of the dividend. Its price-consumption ratio $\hat{v}(x_t, m)$ therefore satisfies the pricing equation $\mathrm{E}[\mathrm{d}(\Lambda\,\hat{v}C)] = 0$ with terminal condition $\hat{v}(x, 0) = 1$, which is Equation (21) with μ_D, σ_D, ρ_{cD}, and ρ_{xD} replaced by $\mu_{ct} + \sigma_{ct}^2/2$, σ_{ct}, 1, and ρ_{cx}.[11] Aggregating the strips across horizons, exactly as in Equation (23) for the stock, gives the wealth claim:

$$N_t = \mathrm{E}_t \int_0^\infty \frac{\Lambda_{t+m}}{\Lambda_t}\, C_{t+m}\, \mathrm{d}m = C_t \int_0^\infty \hat{v}(x_t, m)\, \mathrm{d}m = C_t\, v(x_t), \tag{30}$$

where $v(x) = 1/\kappa(x)$ is the price–consumption ratio of the wealth claim, the analogue of the stock's price–dividend ratio $s(x)$. Section II measures σ_N from dealer equity returns and inverts the translation to recover the consumption volatility schedule.

II. Calibration

The parameters of the model fall into three groups. The first group is set directly. The second, the entire consumption-volatility schedule $\sigma_c(x)$, is measured from the volatility process of the marginal investor. The two remaining parameters are chosen to match eight asset pricing moments, so the model is heavily overidentified. The calibration draws on the intermediary asset pricing literature and identifies marginal investors with financial intermediaries, whose balance sheets are the source of their wealth risk (He and Krishnamurthy 2013; Adrian *et al.* 2014; Haddad and Muir 2021). Time is measured in years throughout.

A. Parameters set directly

The consumption–state correlation is set to $\rho_{cx} = -1$. Under the intermediary interpretation the marginal investor's wealth risk is state risk, driven by a single shock, so wealth and the

[11] Equivalently, by the Feynman-Kac formula, $\hat{v}(x_t, m) = \mathrm{E}_t\big[e^{-\rho m}(C_{t+m}/C_t)^{1-\gamma}\big]$. The current consumption level factors out, so only consumption growth enters the valuation and $\hat{v}$ is a function of the state and the horizon alone.

state are perfectly correlated, and since wealth equals the value of the claim to the consumption stream, consumption tracks wealth and shares that correlation. The interpretation therefore dictates the magnitude of ρ_{cx} but not its sign, which the article takes to be negative. This choice follows the evidence of He *et al.* (2017) that intermediary capital falls when market volatility rises. Since the intermediary's wealth is its equity capital, a rise in the state, which raises volatility, lowers that wealth, so the wealth and state shocks move oppositely, $\mathrm{d}W_{Nt} = -\mathrm{d}W_{xt}$ (that is, $\rho_{Nx} = -1$), and consumption, financed out of that balance sheet, falls with it, giving $\rho_{cx} = -1$. The perfect correlation is consistent with the dealer data (Appendix C). Economically, $\rho_{cx} < 0$ means that positive shocks to the state, which raise consumption volatility, coincide with low consumption, so high-volatility states are high marginal-utility states for the investor. Under the procyclical book leverage of security broker-dealers (Adrian *et al.* 2014; Adrian and Shin 2014), which He *et al.* (2017) contrast with their market-based holding-company measure, the correlation would instead be positive, paired with a falling rate schedule, and Section III.A compares the two.

The relative risk aversion of the marginal investor is set to $\gamma = 2$, a moderate value by asset-pricing standards, so the model does not rely on high risk aversion. The rate of time preference is set to $\rho = 0.01$. The steady-state level of the state variable is normalized to zero without loss of generality, and its diffusion is set to $\sigma_x = 0.2$. This is a normalization of scale, since the volatility and rate schedules are functions of x whose location and width are set in its units. With this value the state variable typically takes values within the interval $(-1, +1)$. Its persistence is set to $\phi = 0.9$, close to the value that related consumption-based models use for their main state variable. The dividend process is also set from the growth of real dividends in Shiller (2015) from 1990 to 2024, which gives a drift of $\mu_D = 3.23\%$ and a diffusion of $\sigma_D = 8.77\%$. The correlations between dividend shocks and both consumption shocks and state-variable shocks are set to zero ($\rho_{cD} = \rho_{xD} = 0$), because the measured correlation between dividend growth and the marginal investor's returns is small and not statistically different from zero at one- and two-year horizons (Appendix A). In addition, shutting the channel down lets the consumption-volatility mechanism operate in isolation.

B. The volatility schedule

Taking the intermediary interpretation to data raises a question that the calibration has to confront. It is not clear what the consumption of a financial intermediary is. Dividends, payouts, and compensation each capture only a part of it, and an institution arguably has no consumption flow in the household sense at all. The calibration does not need to answer this question, because the model links consumption to wealth. The net worth of a publicly traded intermediary is priced continuously in the equity market, so the wealth side is observable, and the model supplies the conversion between the two processes (Section I.F). The volatility schedule measured here should therefore be read as the consumption volatility implied by the observed wealth volatility. The model-derived process can be understood as a consumption equivalent, analogous to the consumption of regular households.

The volatility schedule is measured from the daily value-weighted dealer equity returns of He *et al.* (2017), whose one-year rolling volatility serves as the observable counterpart of the wealth volatility σ_N (Section I.F and Appendix C). The series is an imperfect proxy, since it relies on historical rolling-window estimates, and the equity returns of one group of dealers need not track the wealth of the marginal investor exactly. The measured schedule inherits this measurement error. From 2000 to 2018 this volatility series has mean 23.1%, standard deviation 13.7%, and maximum 72.6% (reached in 2009, in a window spanning the 2008 crisis). The schedule $\sigma_c(x)$ is a capped logistic function of the state (Appendix B), but its parameters are not fitted to the dealer-volatility distribution directly. Dealer volatility measures $\sigma_N(x)$, the volatility of the marginal investor's wealth, which is $\sigma_c(x)$ times a volatility ratio computed inside the model from the consumption–wealth ratio $\kappa(x)$ introduced in Section I.F. The schedule parameters are instead chosen so that the stationary distribution of the model-implied $\sigma_N(x)$[12] matches the empirical 5th, 50th, and 95th percentiles and the mean of dealer volatility. The fit matches all four statistics within one percentage point. Because the volatility ratio depends on the full calibration and the calibration depends on the schedule, the two steps are solved as a joint fixed point, which converges in six iterations (Appendix C). The cap is imposed in the space where the data live, at $\max_x \sigma_N = 0.75$, just above the

[12]The stationary distribution of the state is the distribution it settles into over long horizons. For the process (3) it is normal with mean zero and variance $\sigma_x^2/(-2\log\phi)$. Over a long path, the time average of any function of the state equals its average over this distribution. The stationary distribution of a function of the state, such as $\sigma_N(x)$, is its distribution when x is drawn from it, and the means and percentiles reported below refer to it.

sample maximum. The implied cap on consumption volatility is $\sigma_{\max} = 0.62$. Over the central range of the state the volatility ratio lies between 1.5 and 1.9 and is largest near $x = 0$, so the measured consumption volatility is well below dealer volatility. Figure 1 displays the mapping: the wealth-volatility schedule $\sigma_N(x)$, the recovered consumption-volatility schedule $\sigma_c(x)$, the ratio of the two, and the implied consumption–wealth ratio $\kappa(x)$.

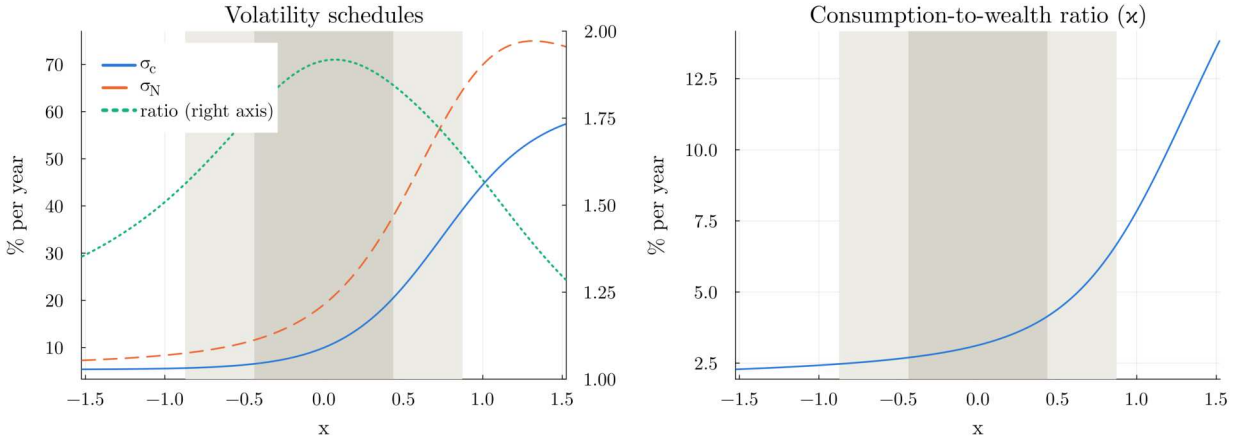


Figure 1: The wealth mapping at the baseline calibration. Left: the wealth-volatility schedule $\sigma_N(x)$, whose stationary distribution is matched to dealer equity volatility, the consumption-volatility schedule $\sigma_c(x)$ recovered from it through the translation (28)–(29), and their ratio σ_N/σ_c (dotted, right axis). Right: the implied consumption–wealth ratio $\kappa(x) = 1/v(x)$. The darker shaded band marks one standard deviation of the stationary distribution of the state either side of zero, and the lighter band two.

C. Parameters chosen to match asset prices

The real-rate schedule $r(x)$ is parameterized directly, as a logistic sharing the location and width of the volatility transition (Appendix B), and the consumption drift $\mu_c(x)$ is then implied by the CRRA Euler equation. There are two free parameters, the minimum and maximum levels of the real rate $r_{\min}$ and $r_{\max}$. They are matched to eight moments: the equity excess return, the mean and the volatility of the P/D ratio, the volatility of returns, the mean real risk-free rate, and the one-year return-predictability coefficient, all from 1990 to 2024 (with prices, dividends, and the CPI from Robert Shiller's online data (Shiller, 2015, 2026), and the market return and the one-month rate from the Fama–French data), and the TIPS 10–5 yr yield spread and excess holding-period return (2004 to 2024, from

the zero-coupon real yield curve of Gürkaynak, Sack and Wright (2010)). The mean real risk-free rate is corrected for the 65-basis-point convenience yield of Treasury bills estimated by Van Binsbergen, Diamond and Grotteria (2022) for maturities below three months from 2004 to 2018, so that it measures the riskless rate that the intermediary itself faces, and the mean equity excess return is measured relative to that rate. These eight moments are the empirical counterparts of the puzzles the model sets out to address jointly: the equity excess return and the mean price–dividend ratio speak to the equity-premium puzzle, the volatility of the price–dividend ratio together with the return volatility to the excess-volatility puzzle, the predictability coefficient to the return-predictability puzzle, the mean real rate to the risk-free-rate puzzle, and the two TIPS moments to the real bond premium puzzle. The article asks whether a reasonable calibration can account for the moments associated with these puzzles. The two parameters are therefore chosen to keep every moment as close as possible to its counterpart in the data. Each deviation is measured in units of the moment's model-implied standard error (SE), the standard deviation of the moment across 2,000 simulated samples of the same length as the data, which is 35 years for the equity and risk-free-rate moments, and the last 21 years of each sample for the two TIPS moments. The two parameters minimize the largest of the eight standardized deviations, so that no moment can be traded away for a closer fit elsewhere. During the search, each model moment is the mean over 200 simulated samples, with the standard errors held fixed (Appendix D.1). The optimum is fully interior. Both $r_{\min}$ and $r_{\max}$ lie strictly inside their bounds, and every optimizer run converges to the same values (reported in Table I and Appendix D.1). Figure 2 shows the criterion over a wide range of the two rates. It has a single valley, whose lowest point is the estimate. The search is also restricted to $r_{\max} > r_{\min}$. With the baseline sign of the consumption–state correlation, a falling schedule would turn real bonds into a hedge, so the two TIPS targets could not be matched.

The procedure is a simulated minimum-distance method in the indirect-inference form of Gouriéroux, Monfort and Renault (1993), which generalizes the simulated method of moments of Duffie and Singleton (1993). It is similar in particular to the partial indirect inference of Dridi, Guay and Renault (2007), in which most parameters of a possibly misspecified model are set directly and a few are estimated on a chosen set of moments, using simulated samples of the same length as the data. It departs from these methods in two respects. It minimizes the largest standardized deviation rather than a quadratic form in all of them, and

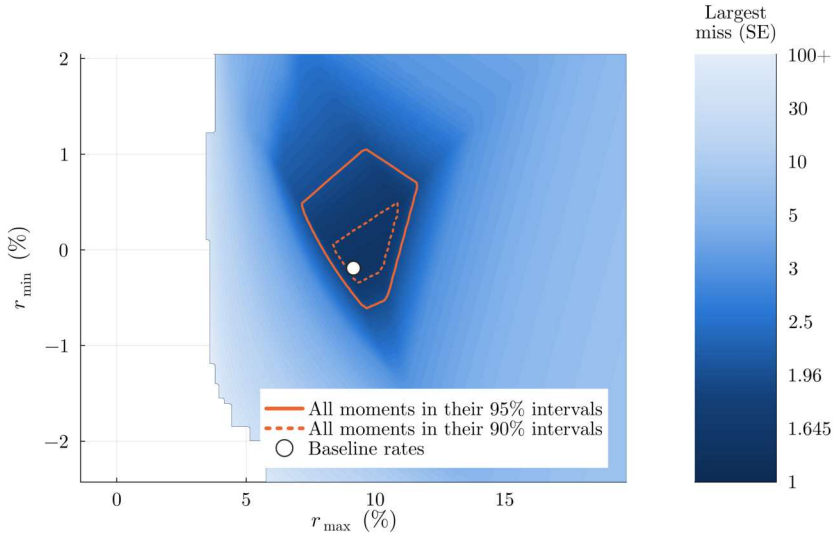


Figure 2: The criterion over the two free parameters. The shading is the largest of the eight deviations between simulated and empirical moments, each in units of its model-implied standard error, and it is darker where the fit is better. The solid and dotted orange lines enclose the rates at which all eight moments lie within their 95% and 90% intervals. Outside the shaded area, asset prices are not well defined. All other parameters are at their baseline values, and every point is evaluated on the same 200 simulated paths as the estimation. The white dot marks the baseline rates.

the measured volatility schedule depends on the two chosen parameters through the volatility ratio, so the measurement and the choice of the parameters are iterated to a joint fixed point.

III. Results

A. Overall fit

The model matches all eight targeted moments with only two parameters chosen to fit them. Table II compares the model with the data, grouping the eight moments by the puzzle each one corresponds to. Each of the eight moments lies within one and a half standard errors of its target, and the largest deviation is that of the mean P/D ratio, at 1.4 standard errors. Figure 3 shows how the moments respond when one rate moves away from its estimate while

Table I: Baseline parameters. The three panels correspond to the three groups of parameters described in Section II: set directly, measured from the volatility of NY Fed primary-dealer equity returns through the wealth mapping, and chosen to match eight asset-price moments.

Parameter	Symbol	Value
Set directly		
Consumption–state correlation	ρ_{cx}	-1
State-variable persistence	ϕ	0.90
Time preference	ρ	0.010
Risk aversion	γ	2
State-variable diffusion	σ_x	0.20
Dividend drift	μ_D	0.032
Dividend volatility	σ_D	0.088
Measured (via the wealth mapping)		
Consumption vol. (floor)	$\sigma_{\min}$	0.053
Consumption vol. (cap)	$\sigma_{\max}$	0.615
Volatility transition (center)	x^*	0.744
Volatility transition (width)	w	0.308
Chosen to match (8 moments, 2 parameters)		
Real rate (low volatility)	$r_{\min}$	-0.002
Real rate (high volatility)	$r_{\max}$	0.091

the other is held fixed. In the criterion of the search, which uses the standard errors frozen for the estimation (Appendix D.1), three moments bind at the estimate, each at 1.46: the equity premium, which lies below its target, and the mean P/D ratio and the volatility of returns, which lie above theirs. Lowering $r_{\min}$ raises the premium but also the volatility of returns, and raising $r_{\max}$ lowers the P/D ratio but also raises the volatility of returns, so neither rate can move without pushing one of these moments further from its target.

Two further checks support the fit. The first is a joint test of the eight moments, which compares the largest of the eight deviations in the data with the largest deviation in each of the model's simulated samples. In 56% of those samples it is greater than in the data, so the deviations of the moments from their targets look like those of a typical sample from the model (Appendix D.1). The second is a leave-one-out exercise, which drops each target in

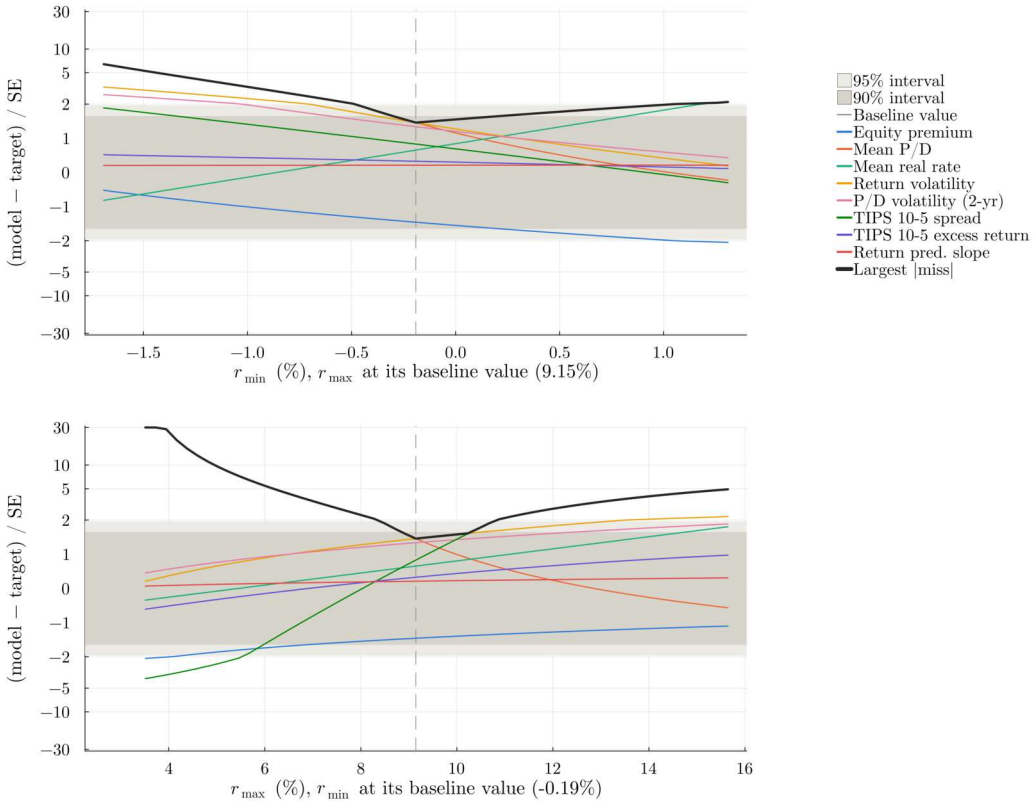


Figure 3: Standardized deviation of each targeted moment from its target, (model − target)/SE, with the standard errors frozen for the search (Appendix D.1), as one rate varies with the other held at its baseline value: $r_{\min}$ in the upper panel and $r_{\max}$ in the lower panel. The vertical axis is linear within ±2 standard errors and logarithmic beyond. The shaded bands mark the 90% and 95% intervals, the black line is the largest absolute deviation, and the dashed line marks the baseline value.

Table II: Simulated moments versus empirical targets. The targets and their sources are described in Section II. The Model column reports the mean across 2,000 independent simulated paths of $T = 35$ years, with the TIPS moments over the last 21 years of each. SE is the model-implied standard error defined in Section II. The risk-free-rate target and the equity excess return are adjusted for the convenience yield of bills (Section II). A moment counts as matched when its target lies within the model's 95% interval, 1.96 standard errors around the Model entry, as all eight do here.

Puzzle	Moment	Target	Model	SE
Equity premium	Equity excess return ($\bar{r}_e$, %)	8.088	5.299	2.188
	Mean P/D ratio ($\overline{P/D}$)	53.85	77.65	16.74
Risk-free rate	Mean real risk-free rate ($\bar{r}_f$, %)	0.670	1.154	0.919
Excess volatility	Std. stock return (σ_e, %)	15.47	19.95	3.38
	Median 2-yr std $\log(P/D)$ ($\operatorname{med}\sigma^{2y}_{pd}$)	0.0671	0.0931	0.0207
Return predictability	Return pred. slope ($\hat{\beta}_1$)	0.215	0.237	0.128
Real bond premium	TIPS 10–5 yr excess return (TIPS rx., %)	0.709	0.924	0.525
	TIPS 10–5 yr yield spread (TIPS sp., %)	0.457	0.507	0.064

turn and chooses the two rates again to match the remaining seven. In all eight cases, every remaining moment stays within its 95% interval. The one target that the other seven do not pin down is the mean P/D ratio, which rises to 195.0 when it is left out (Appendix D.2).

The fit does depend on the sign of the consumption–state correlation. The opposite pairing of Section II, a positive correlation with a falling rate schedule, measured from the same dealer data, matches five of the eight moments, including the risk-free rate and both bond moments. Its equity premium counts among them only because it varies widely across that model's samples, as its mean of 3.1% is far below the 8.1% in the data. It fits worse overall, missing the volatility of returns and the level and volatility of the P/D ratio (Appendix D.5). In general, the two views of intermediary leverage need not conflict, since Kargar (2021) reconciles them in a single model, with leverage procyclical among security broker-dealers and countercyclical at the holding-company level, where the equity data used here are measured.

B. Consumption dynamics

Figure 4 shows the main model-implied functions of the state variable x, together with its

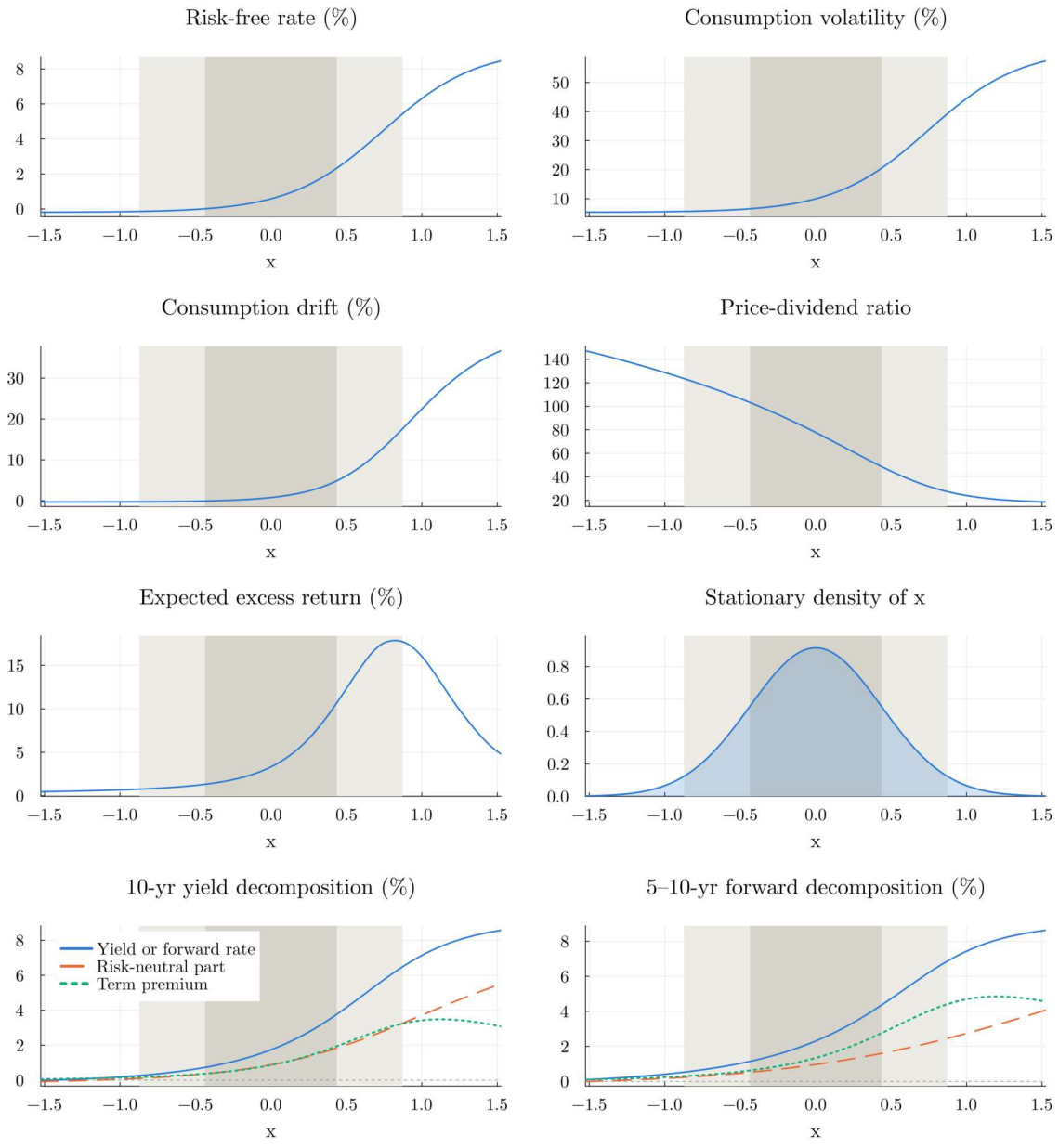


Figure 4: Model-implied quantities under the baseline calibration. Risk-free rate $r(x)$, consumption volatility $\sigma_c(x)$, consumption drift $\mu_c(x)$, price–dividend ratio $P/D(x)$, expected equity excess return $\mathbb{E}[R_e - r](x)$, stationary distribution of x, and decompositions of the 10-year real bond yield and of the 5–10-year real forward rate into their risk-neutral parts and the term premium. The darker shaded band marks one standard deviation of the stationary distribution of x either side of the steady state, and the lighter band two.

stationary distribution and the decompositions of the 10-year real yield and the 5–10-year real forward rate. Consumption volatility $\sigma_c(x)$, recovered from dealer volatility through the wealth mapping, increases monotonically with the state variable and runs from 5.8% at the 5th percentile through 10.0% at the median to 32.2% at the 95th, with an unconditional mean of 13.2%. This is well below dealer-return volatility, because part of that volatility reflects movements in the consumption–wealth ratio (Section I.F), but still roughly an order of magnitude above aggregate consumption volatility for a representative household. Consumption drift $\mu_c(x)$, implied by the rate schedule, increases monotonically, from slightly negative values in low-volatility states to 0.8% at the median and 12% at the 95th percentile of the state, with a stationary mean of 2.6%. The drift and the volatility of consumption therefore rise together, a feature that many models in the literature share (Section IV).

C. The real rate

The risk-free rate stays within 0.8 percentage points of its floor of -0.2% over the entire lower half of the stationary distribution, passes through $+0.6\%$ at the stationary median, and rises materially only in the high-volatility tail: $+4.3\%$ at the 95th percentile of the stationary distribution and $+6.4\%$ at the 99th. The model therefore does not predict rising real rates when volatility is low. In the high-volatility tail, the rising rate corresponds to a consumption drift that rises with the state. Through the Euler equation the real rate combines an intertemporal substitution term in the expected growth of consumption with a precautionary term in its volatility, and it rises with the state only where the growth term rises faster than the precautionary term.

The model also does not generate a risk-free-rate puzzle. Its mean of 1.2% lies about half a standard error above the target of 0.7%, and its time-series variation is tame, as the sample simulation of Internet Appendix IA.A shows. The target includes the correction for the convenience yield of Treasury bills (Section II). If anything, the correction is conservative, because beyond the convenience yield an intermediary must also earn more than its financing rate to hold a Treasury position. Indeed, Fleckenstein and Longstaff (2020) measure this cost in the Treasury note futures market, as the gap between the financing rate implied by the futures price and the repo rate at which the notes themselves are financed. They interpret the gap as the cost of renting intermediary balance sheet space, and find that it averages

about 59 basis points from 1991 to 2018.[13] Appendix D.4 re-runs the calibration without the correction. Even then, the model's rate lies about one standard error above the near-zero uncorrected target, and all eight moments remain matched.

D. Equity premium, excess volatility, and return predictability

The P/D ratio declines steeply with x. The stock market is highly valued when the state variable is low (consumption volatility low, discount rates low) and cheaply valued when x is high. On average, the model implies a P/D ratio of 77.7, against 53.8 in the data, so it tends toward higher valuations than those observed. The expected equity excess return correspondingly rises with x over the bulk of the stationary distribution and peaks in high-volatility states, beyond which the P/D ratio flattens and returns become less exposed to the state. On average, the model delivers an equity premium of 5.3%, against 8.1% in the data. For comparison, Mehra and Prescott (1985) report that the largest premium obtainable in their model is 0.35% with risk aversion as high as ten.

The model naturally produces excess volatility in stock prices relative to dividends, in the sense that the variance of stock prices exceeds what could be justified by the present value of future dividends computed with a constant discount rate, as documented by Shiller (1981). This excess volatility is a direct consequence of time-varying discount rates driven by the state variable. Even with stable dividend fundamentals, stock prices fluctuate substantially. Quantitatively, the model generates a standard deviation of log real stock returns of 19.9% and a median standard deviation of $\log(P/D)$ within non-overlapping 2-year windows of 0.093, somewhat above the 15.5% and 0.067 in the sample from 1990 to 2024 (Table II).[14]

Because the equity premium is state-dependent and the P/D ratio is a monotone function of the same state variable, current valuation ratios predict future returns, as documented

[13]The spread is far from constant. It reached 200 basis points or more during the LTCM crisis of 1998, the Treasury buybacks from 2000 to 2002, and the financial crisis of 2008, and it rises significantly with the VIX (Fleckenstein and Longstaff 2020). The gap between the rate an intermediary faces and the observed rate may therefore be largest in high-volatility states, where the model's rate rises.

[14]The full-sample standard deviation of $\log(P/D)$ in the data (0.27 from 1990 to 2024 and 0.48 from 1926 to 2024) is likely inflated by slow-moving shifts in the long-run level of the P/D ratio, driven, for instance, by gradual changes in risk aversion, trend growth, or the equity risk premium over decades. Such level shifts would show up as a large standard deviation even if short-run fluctuations were modest. The standard deviation within 2-year windows captures local variability around whatever the prevailing level happens to be, and is therefore a cleaner target for a model with fixed structural parameters.

empirically by Campbell and Shiller (1988). Table III reports the slopes from this predictive regression. The model implies slopes of 0.24, 0.72, and 1.04 at horizons of one, four, and eight years, capturing the well-documented pattern of predictability increasing with horizon. Only the one-year slope is targeted, and the longer-horizon slopes are close to the estimates of 0.83 and 1.32 from 1990 to 2024, given their dispersion across the model's samples. The slopes are much smaller from 1926 to 2024, at 0.06, 0.21, and 0.37, so the strength of predictability in the data depends on the sample. The empirical standard errors of the long-horizon slopes are small, but with overlapping returns in a 35-year sample they are themselves unreliable, so the evidence from these regressions is imprecise.

Table III: Return predictability regressions: OLS of the cumulative log real stock return $r_{t\to t+T}$ on the log dividend-price ratio $\log(D_t/P_t)$ at horizons $T = 1, 4, 8$ years, $r_{t\to t+T} = \alpha_T + \beta_T \log(D_t/P_t) + \varepsilon_{t+T}$. A positive slope β_T indicates that a cheap market (low P/D ratio) predicts high future returns, consistent with time-varying discount rates. Empirical standard errors (in parentheses) are Newey–West with bandwidth equal to the horizon in months. Model standard errors are the cross-simulation standard deviation of $\hat{\beta}_T$. Model R^2 is the average in-sample R^2 across simulations.

	Model			1990–2024			1926–2024		
Horizon	β	(s.e.)	R^2	β	(NW s.e.)	R^2	β	(NW s.e.)	R^2
1 year	0.237	(0.128)	0.109	0.215	(0.071)	0.134	0.061	(0.037)	0.023
4 years	0.716	(0.332)	0.318	0.831	(0.230)	0.454	0.209	(0.096)	0.079
8 years	1.036	(0.484)	0.443	1.316	(0.114)	0.763	0.372	(0.159)	0.149

E. Bond term premia and the yield curve

Generating a positive and sizeable bond term premium is one of the central challenges of asset pricing models (Backus *et al.* 1989), and the challenge is even greater for real term premia. In the present model, long-term bonds are risky. In bad states (high x), consumption volatility is high and consumption is expected to recover, so the consumption drift is high and, through the Euler equation, so is the real rate. Long-term bonds therefore lose value when x rises, which is when the marginal investor's marginal utility is high, and investors require a positive real term premium to hold them. The mechanism relies on the rate and volatility schedules sharing the same transition in the state, a restriction examined in Appendix D.3.

The yield decomposition in Figure 4 shows that the 10-year real yield and the term premium both increase with x. At low values of x, the term premium is near zero and the yield is low. In the central range of the stationary distribution, the term premium is positive and increasing, reflecting that long-term bonds are risky precisely when the marginal investor's consumption volatility is high. At very high x the total yield rises steeply, with the risk-neutral yield contributing somewhat more than the term premium as expected short rates become very high.

The two targeted bond moments are the TIPS 10–5 yr yield spread and the TIPS 10–5 yr excess holding-period return, the monthly log return difference between the two bonds expressed at an annual rate (Appendix A). The spread and the excess return are used rather than yield levels because a liquidity premium that affects the overall level of TIPS yields largely cancels in the difference between two maturities, and the model, in which all assets are perfectly liquid, has no such premium. The model generates a positive spread and a positive excess return, at 0.51% against 0.46% in the data for the spread and 0.92% against 0.71% for the excess return. These moments pin down the slope of the real yield curve, but not how yields divide between the risk-neutral yield (RNY, Equation (14)) and the term premium (TP, Equation (13)). Most estimates of this division in the literature find a positive real term premium, but they differ on its size and on its shape across maturities.[15] In the model, the 10-year term premium averages 1.10% and is about half of the 10-year yield across most of the stationary distribution. The 5–10-year forward term premium ranges from about 0.4% at the 5th percentile of the stationary distribution to about 4% at the 95th, close to the range from around zero to about 4% estimated by Abrahams *et al.* (2016, their Figure 5) from 1999 to 2014, although they attribute nearly the entire forward rate to the term premium, a larger share than in the model. The estimates of Christensen and Rudebusch (2019) are lower on average, mostly between about −0.7% and 3% from 1998 to 2016. The model's term premium is also close to the averages estimated by Haubrich *et al.* (2012) from 1982 to 2010, at 0.23%, 0.57%, and 1.10% for two, five, and ten years, against their 0.25%, 0.57%, and 1.03%. The estimates of d'Amico *et al.* (2018) are smaller than these and less variable. In the series that Federal Reserve Board staff maintain (Kim *et al.*, 2019), the 10-year premium

[15] An exception is Ang, Bekaert and Wei (2008), who find an essentially flat real yield curve and hence little real term premium. They do not use TIPS data, but infer real rates from nominal yields and inflation from 1952 to 2004.

averages 0.4% from 2004 to 2024 and the 5–10-year forward premium lies between −0.4% and 1.8%. The model therefore lies within the range of the published estimates, toward its upper end. Its real yields are higher on average than in the TIPS data, partly because its short rate is the rate the intermediary faces rather than the Treasury rate. The estimated premium also varies with risk, as in the model. Christensen and Rudebusch (2019) find a countercyclical forward premium that is elevated in recessions. In the series of Kim *et al.* (2019), the month-end 10-year and 5–10-year forward premia both reach their highest value from 2004 to 2024 in October 2008, at the height of the crisis. Haubrich *et al.* (2012) find that real risk premia rise with the volatility of the short real rate at short maturities and with the volatility of the central tendency of inflation at longer ones.

Internet Appendix IA.B reports bond return predictability regressions following Fama and Bliss (1987) (FB) and Campbell and Shiller (1991) (CS). In the model, as in the data, the slopes depart in the same direction from the values the expectations hypothesis implies, zero for FB and one for CS. FB slopes are positive and above one at maturities of two to five years, exceeding the empirical values by about 1.0 to 1.6, and CS slopes are negative and steeper than the data at every maturity. They vary so widely across 35-year samples, however, that the empirical values lie within 1.25 standard deviations of the model's means at every maturity. The regressions are therefore weak evidence on the real term premium, since their slopes are imprecise in samples of this length and the empirical values were estimated for nominal bonds. The TIPS moments and the estimates of the real term premium discussed above are more informative.[16]

F. Asset prices from the recovered state

The analysis so far has focused on unconditional moments. This subsection asks a different question, namely how well the model captures the state of the economy over time, which tests its conditional predictions. Inverting the dealer-volatility series through the wealth mapping $\sigma_N(x)$ recovers the state variable x_t at each date, and the model-implied prices at that state

[16]The econometric literature documents severe small-sample distortions in predictive regressions in the term structure (Bekaert, Hodrick and Marshall 1997; Bauer and Hamilton 2018). The simulations here point to a further problem. Even when the model is the true data-generating process, the slope estimates from 35-year samples are so dispersed (Tables IA.1 and IA.2) that the regressions may have little power to distinguish between models with very different term premium dynamics. A systematic investigation is left for future research.

can be compared with the observed ones. The series is the one-year rolling volatility of the daily dealer equity returns in the data of He *et al.* (2017) (Section II), a variable constructed here rather than one of their measures, such as the intermediary capital risk factor. The volatility schedule is measured from this series alone, and the calibration compares the model with the data only through unconditional moments, so nothing in the calibration aligns the model-implied prices with the observed ones over time. The model-implied prices comove positively with the data. The model-implied price–dividend ratio has a levels correlation of 0.30 with the observed ratio (0.39 in changes), and the model-implied 10–5 yr TIPS spread a levels correlation of 0.50 (0.15 in changes). A single measured state therefore carries common variation in both equity valuations and the slope of the real term structure. The correlations are well short of one, which is expected for a one-factor model whose state is measured from the volatility of dealer returns alone, without any price level, valuation ratio or yield. The model-implied 10-year TIPS yield, by contrast, is essentially uncorrelated with the observed one, at 0.04 in levels and 0.03 in changes. Appendix E reports the full comparison.

G. Remaining challenges

The model matches the moments that define each of the puzzles studied here, but it remains stylized, and some relations among these moments, and some features of asset prices beyond them, are harder for it to reproduce. The clearest tension concerns the Sharpe ratio of equities. With power utility, the Sharpe ratio of any asset is bounded by the volatility of the stochastic discount factor, approximately $\gamma\sigma_c$ (Hansen and Jagannathan 1991; Lettau and Uhlig 2002). With the volatility of aggregate consumption and a risk aversion of two, the bound is below 0.1, against an observed Sharpe ratio of 0.52. The high consumption volatility of the marginal investor raises the model's Sharpe ratio to 0.27 on average, about half the observed value.[17] The premium and the volatility of returns each lie within their 95% intervals, but the premium falls short of its target while the volatility exceeds its own, so their ratio misses by more. Taken together, the observed premium and volatility are consistent with the model, since 14% of its 35-year samples lie farther from the model's means, with

[17]The model's conditional Sharpe ratio is countercyclical, rising from 0.08 at the 5th percentile of the stationary distribution of x to 0.61 at the 95th. Its variation is similar in size to that of the habit model of Campbell and Cochrane (1999) and smaller than that estimated by Lettau and Ludvigson (2010), who also find the conditional mean and volatility of returns moving in opposite directions, whereas in the model they rise together over most of the state space.

the distance measured relative to the covariance of the two moments across samples. Their ratio is not, since only 1.9% of the samples reach the observed Sharpe ratio, just outside the model's 95% range (Figure D1 in Appendix D.6). The model's Sharpe ratio is close to its bound, so closing the gap requires a higher price of risk, whether through more volatile consumption or an additional channel.

The Sharpe ratio also limits how far the premium can rise. Within the model, the two rates move the premium and the volatility of returns together, and the Sharpe ratio stays at about 0.3 or below. With all other moments inside their 95% intervals, the premium rises to at most about 7%, at which point the mean P/D ratio and the TIPS spread reach the edges of their intervals. Raising the premium to its observed value of 8.1% in this way would take a return volatility of about 28%, against 15.5% in the data (Appendix D.6).

A related tension appears in valuations. Although the mean P/D ratio lies within its interval and the joint test of the eight moments does not reject the model, the model ties low valuations to a high risk-free rate and high return volatility. Across its samples, the mean P/D ratio has a correlation of -0.93 with each of the two. In the tenth of samples with the lowest valuations, the P/D ratio averages 47.8, close to the 53.8 in the data, but the risk-free rate averages 3.1% and the volatility of returns 24.6%, against 0.7% and 15.5% in the data, while the premium stays at about 5.0%. The Sharpe ratio of these samples is therefore even lower, at 0.20. In the data, low valuations come with a low risk-free rate and moderate volatility, a combination the model does not produce. Among its samples with a mean P/D ratio of 60 or less, the lowest risk-free rate is 1.7% and the lowest volatility of returns 18.8%, against 0.7% and 15.5% in the data. Across all 28 pairs of targeted moments, the data lie outside the model's 95% region in only three, each pairing the mean P/D ratio with the risk-free rate, the volatility of returns, or the volatility of the P/D ratio (Figures D2.A and D2.B in Appendix D.6).

The single state variable also limits the model mechanically, since it moves all prices together. The state recovered from dealer volatility tracks the P/D ratio and the TIPS spread only in part, and is essentially uncorrelated with the level of the 10-year TIPS yield (Section III.F). Section IV discusses extensions that could address these issues, such as a second state variable and mechanisms that raise the price of risk.

IV. Discussion

The exogenous process introduced in this article distills a mechanism that many articles in the literature share. The process has two main elements. The first is high volatility in the consumption of the marginal investor. The second is the interaction between drift and diffusion, taken up below. High volatility is widespread, including in work that does not address the term premium, and the literature gives several reasons why it is not visible in aggregate data. The consumption of wealthy households is more volatile and more exposed to stock returns than the aggregate (Ait-Sahalia *et al.* 2004). In some articles the volatility is idiosyncratic. Income shocks are uninsurable, so individual consumption is volatile while the cross-sectional average is smooth, and every agent is marginal and every Euler equation holds (Constantinides and Duffie 1996), with Mitra and Xu (2024) adding labor adjustment costs that arise from search frictions. In other articles, only a subset of agents have high volatility, and the remaining households are kept out of the markets where risk is priced, whether by information costs (Basak and Cuoco 1998), by a capital constraint limiting how much household wealth reaches risky assets (He and Krishnamurthy 2013), or by clienteles whose maturity demands leave arbitrageurs to price long bonds (Vayanos and Vila 2021; Kekre, Lenel and Mainardi 2024). In Schneider (2026) the volatility is concentrated in rare crises, so it barely shows in average aggregate data.

Apart from high volatility, many models, especially those that address the real term premium, also share the interaction between drift and diffusion (Vayanos and Vila 2021; Schneider 2022; Mitra and Xu 2024; Kekre *et al.* 2024; Schneider 2026). In all of them, the drift, or expected growth, of consumption or wealth rises with its volatility. Where the real rate is determined within the model, this pushes it up in the high-volatility states. The feature holds even for the habit model of Wachter (2006), once consumption is taken to be the habit-adjusted consumption that enters marginal utility.[18] In models written in terms of wealth, the link arises because investors require higher returns when volatility is higher. In models written in terms of consumption, it arises because consumption is expected to recover from the states where volatility is high. The results of this article suggest that it is this structure, combined with high volatility, that accounts for the puzzles, the real bond

[18]Habit models admit two interpretations, depending on whether the habit is incorporated in consumption or in risk aversion. On the first, habit-adjusted consumption is volatile while risk aversion is constant. On the second, consumption is smooth while effective risk aversion is high and time-varying.

premium among them. The precise form of the mechanism differs across models, and not all articles report the level of volatility that their marginal investor faces, so it is not clear which of them succeeds with the least volatility.[19] Further research could combine these and other mechanisms within models with high-volatility investors, and assess how much volatility each requires and how closely each accounts for the facts.

The model's limits, set out in Section III.G, point to natural extensions. Its Sharpe ratio is about half the observed value, and it ties low valuations to a high risk-free rate and high return volatility. A second state variable could separate valuations from the cycle in rates and volatility, let the level and the slope of the yield curve move partly independently, as they do in the data, and capture variation in the real rate that is unrelated to the volatility cycle, such as a slow-moving component of expected growth. Rare disasters (Rietz 1988; Barro 2006; Wachter 2013), or recursive preferences as in Bansal and Yaron (2004), which would also price the persistent movements in expected growth already present in the model, could raise the price of risk further and close the gap in the Sharpe ratio. Neither delivers a positive real term premium by itself, however. In Wachter (2013) the real yield curve slopes downward, and with a preference for early resolution of uncertainty recursive preferences tend to lower the real term premium (Internet Appendix IA.C). A model extended in these ways, in particular with a second state variable, could also serve as a lens on historical episodes. The prices of the present model at the state recovered from dealer volatility already move partly with price–dividend ratios and TIPS spreads (Section III.F), but not with the level of real yields, which a second state could capture. A richer model, with its states recovered from the data in the same way, could then shed light on episodes such as the 2008 financial crisis, the post-crisis period of compressed term premia and the 2022 rate cycle.

V. Conclusion

This article has shown that a parsimonious model, with a single state variable and two free

[19] Schneider (2026) combines several features that together generate a positive real term premium. Savers, with time-separable preferences and a risk aversion of five, are the marginal investors in long-term real bonds, and their consumption volatility is high in infrequent crises in which the intermediaries' financing constraint binds. Since the economy spends most of its time in a normal regime with moderate asset-price volatility, its average consumption volatility is presumably not too high, while the average term premium is positive and rises with maturity.

parameters, can address five classic asset pricing puzzles together: the equity premium, the risk-free rate, the real bond premium, excess volatility and return predictability. In the model, the marginal investor faces high and time-varying consumption volatility, measured from dealer returns in the spirit of He *et al.* (2017), and the model matches each of the eight targeted moments that define these puzzles. In particular, it generates a positive and sizeable real term premium where standard frameworks predict one that is near zero and negative. Even recursive preferences, which can match the equity premium, tend to lower the real bond premium rather than make it positive (Internet Appendix IA.C). Based on this analysis, the high and time-varying consumption volatility of the marginal investor appears to be an essential ingredient in this type of model.

In retrospect, the solution may have been in plain sight. The equity premium puzzle and the puzzles alongside it have long signaled that consumption volatility would need to be far higher than in aggregate data to rationalize observed risk premia. Rather than a symptom of model failure, this can be read as a sign that the consumption volatility of the marginal investor is genuinely high and time-varying. It is then unsurprising that many strands of the literature reach a similar conclusion, that the marginal investor is not the median household.

This reading matters for regular households. Their consumption is relatively stable, so the high risk premia in financial markets pay far more than their own risk would demand. This makes limited participation by regular households, or by other institutions not exposed to high volatility, a potential missed opportunity, with implications for wealth accumulation and for how asset returns are distributed across the population. The model does not attempt to explain whether and why participation is limited, or why regular households do not otherwise influence the prices set in these markets, and these questions deserve further research in connection with the high volatility of marginal investors. If, as financial services improve, households participate more directly in equity and bond markets, the risk profile of the marginal investor should move closer to that of the aggregate, and both the equity premium and the real term premium would be expected to compress.

Appendix A. Empirical Moments

The main empirical moments are reported in Tables II and III in the main text. Table AI below extends these by reporting moments across multiple sample periods for comparison. The TIPS yields are zero-coupon yields computed from the Nelson–Siegel–Svensson parameters of the curve of Gürkaynak *et al.* (2010) at every maturity, including the fractional ones that the holding-period returns require, and excess returns are monthly log holding-period returns, annualized.

The dividend process (15) is set from the growth of real dividends in Robert Shiller's online data (Shiller, 2015, 2026) from 1990 to 2024. Because $\log D_t$ has independent increments, σ_D is the standard deviation of log dividend growth at an annual horizon, measured directly at that horizon. It cannot be recovered by scaling up monthly growth, since the monthly dividend series interpolates four-quarter totals, so monthly growth is close to deterministic and strongly serially correlated. The annual dividend is the total paid over the year, and for a random walk the growth of such totals has about two thirds of the variance of the growth between points in time (Working, 1960). The exact ratio for a year divided into m periods is $(2m^2+1)/(3m^2)$, 0.67 for months, so scaling the measured standard deviation of 7.06% by $\sqrt{3/2}$ gives 8.64%. This scaling holds for the population. In a 35-year sample the standard deviation is lower on average, because the growth rates of annual totals are serially correlated. The diffusion is therefore set to $\sigma_D = 8.77\%$, the value at which the model's dividends, aggregated in the same way, have a standard deviation of annual growth of 7.06% on average across 35-year samples. The drift is set to $\mu_D = 3.23\%$, the mean log growth of 2.84% plus $\sigma_D^2/2$. On the model's simulated paths, dividends then grow at a mean of 2.87% with a standard deviation of 7.09% a year, against 2.84% and 7.06% in the data.

The correlations of dividend shocks with consumption and state shocks are set to zero (Section II). Over non-overlapping windows of one and two years, December 2000 to November 2018, the correlation between real dividend growth and the log returns of the dealer portfolio of He *et al.* (2017) is -0.17 and -0.07, with 18 and 9 observations, and neither is distinguishable from zero. Shorter horizons are not informative, because the monthly dividends of Shiller (2015) are four-quarter totals interpolated to months.

Table AI: Stock market and dividend moments: model and data. Shiller prices, dividends, CPI and total-return index, with the Fama–French one-month rate. Returns and rates are annualized. Dividend growth is real and annual. Its mean is the one behind μ_D, and its standard deviation is the counterpart of σ_D before the time-aggregation correction. The equity premium in the data is measured over the bill rate, before the convenience-yield correction of Section II, which lowers the 1990–2024 target to 8.09%.

Moment	Model	1990–2024	1999–2024	1926–2024
Mean P/D ratio	77.7	53.8	58.0	34.5
Median 2-yr std $\log(P/D)$	0.093	0.067	0.074	0.096
Std. $\log(P/D)$	0.287	0.271	0.206	0.483
Mean annual real div. growth (%)	2.87	2.84	3.37	1.98
Std. annual real div. growth (%)	7.09	7.06	8.06	10.43
Mean real stock return (%)	6.45	8.76	6.94	8.55
Std. log real stock return (%)	19.95	15.47	15.90	18.46
Mean equity premium (%)	5.30	8.74	7.58	8.24

Appendix B. Functional Forms for Consumption Dynamics

Log consumption follows the diffusion

$$\mathrm{d}\log C_t = \mu_c(x_t)\,\mathrm{d}t + \sigma_c(x_t)\,\mathrm{d}W_{ct}, \tag{B1}$$

where x_t is the scalar state variable and W_{ct} a standard Brownian motion (Section I).

The volatility schedule is a logistic function of the state,

$$\sigma_c(x) = \sigma_{\min} + (\sigma_{\max} - \sigma_{\min})\,L\left(\frac{x - x^*}{w}\right), \qquad L(u) = \frac{1}{1 + e^{-u}}, \tag{B2}$$

which rises strictly from the floor $\sigma_{\min}$ to the cap $\sigma_{\max}$, so high-x states are high-volatility states. The floor and the cap keep the risk-adjusted dynamics well behaved at extreme states. In particular, the cap bounds the drift of the state in the strip PDE (21), which the convergence of the price–dividend ratio requires.

The real-rate schedule is a logistic with the same location x^* and width w as the volatility schedule (B2),

$$r(x) = r_{\min} + (r_{\max} - r_{\min})\,L\left(\frac{x - x^*}{w}\right), \tag{B3}$$

where $r_{\min}$ and $r_{\max}$ are the two parameters chosen to match asset prices (Section II). The rate is therefore monotone by construction, flat near $r_{\min}$ in low-volatility states and rising through the same band of states over which volatility rises. The consumption drift is then implied by the CRRA short-rate identity rather than parameterized separately:

$$\mu_c(x) = \frac{r(x) - \rho}{\gamma} + \frac{\gamma}{2}\,\sigma_c(x)^2. \tag{B4}$$

Appendix D.3 examines the restriction that the two schedules share their location and width.

Appendix C. Intermediary Consumption Volatility

This appendix section documents two measurements used in Section II: the consumption volatility schedule recovered from dealer equity volatility, and a check of the assumed perfect correlation $\rho_{cx} = -1$ against the correlation between dealer returns and dealer volatility.

The volatility schedule. The data are the daily value-weighted equity returns of NY Fed primary dealer holding companies (He *et al.*, 2017), from December 2000 to November 2018 (216 months), summarized by their one-year rolling volatility (Panel A of Table EI).[20] The persistence of the state is set directly to $\phi = 0.9$ (Section II), and the monthly AR(1) coefficient of the rolling volatility series, which implies an annual persistence of 0.891, serves as a check. Given ϕ, the parameters $(\sigma_{\min}, x^*, w)$ of the logistic schedule (B2) are chosen to minimize the sum of squared relative deviations of the 5th, 50th and 95th percentiles and the mean of the model-implied stationary σ_N from their dealer-volatility counterparts, and the cap is set so that the largest σ_N equals 0.75. Because the volatility ratio depends on the state and the calibrated parameters depend in turn on the schedule, the two steps form a joint fixed point. The volatility ratio is computed at the current calibration, the schedule is refitted, the two free parameters are re-estimated, and the cycle repeats until the schedule settles. This takes six iterations, with the schedule parameters changing by less than 0.005 in the last one. The resulting fit is close at every targeted point of the σ_N distribution: 5th percentile 9.5% (data 9.6%), median 19.1% (data 19.5%), 95th percentile 55.6% (data 56.5%), mean 23.9% (data 23.1%). The dealer series is a one-year rolling estimate, while the schedule is fitted to the instantaneous volatility σ_N. Applied to wealth returns simulated from the model, the same rolling estimator gives quantiles within about 0.5 percentage points of those of σ_N, so the difference does not matter for the measurement.

With $\rho_{cx} = \rho_{Nx} = -1$, Equation (29) gives $\sigma_N = \sigma_c + (\kappa'/\kappa)\,\sigma_x$, so the volatility ratio σ_N/σ_c is $1 + (\kappa'/\kappa)\,\sigma_x/\sigma_c$. It exceeds one because κ increases with the state, so the model's step of the measurement only ever removes volatility. It is largest near the middle of the stationary distribution, where the state spends most of its time, and lies between 1.51 and 1.92

[20] Specifically, the `intermediary_value_weighted_investment_return` series from the He *et al.* (2017) dataset is used. This is the value-weighted investment return to a portfolio of NY Fed primary dealers' publicly-traded holding companies. Unlike the intermediary capital risk factor, this portfolio is tradable. Per the dataset's documentation, He, Kelly, and Manela source the daily series entirely from Bloomberg.

for x between -1 and 1 (Figure 1). Because the ratio exceeds one in every state, including the high-volatility states that generate the equity and bond premia, relying on the model's volatility ratio works against the volatility mechanism rather than in its favor.

The return–volatility correlation. The calibration assumes a perfect negative correlation between dealer returns and innovations to dealer volatility, $\rho_{Nx} = -1$, which under the wealth mapping corresponds to $\rho_{cx} = -1$ (Equation (29)). At first glance this correlation looks weak in the data. Computed from one-month returns and volatility changes, it is only -0.22. It grows steadily over longer windows, however, and reaches -0.73 at twelve months (Table CI). Short-horizon estimates are biased toward zero, because the series is a one-year rolling estimate and is therefore an imperfect measure of volatility over short windows.

To gauge this bias, the same estimator is applied to wealth returns simulated from the model (the model counterpart of the dealer series, combining consumption shocks with the valuation term $\mathrm{d}\log v(x_t)$) at two values of the underlying correlation. The bias is large. Even at the perfect-correlation limit, the simulated estimator reaches only about half of the true correlation at twelve months, and it moves little with the underlying ρ_{cx} relative to its sampling dispersion. The measured correlations are stronger than the simulated ones at every horizon, but within 1.1 to 1.9 standard deviations of them at $\rho_{cx} = -1$, and a weaker correlation, such as $\rho_{cx} = -0.3$, leaves them further away. The value $\rho_{cx} = -1$ therefore does not overstate the measured correlations. What remains of the gap may reflect sources of comovement, such as jumps, that a correlated diffusion lacks. Reversing the sign of ρ_{cx} also reverses the sign of the simulated correlations (Appendix D.5).

Appendix D. Calibration Procedure and Robustness

This appendix section documents the numerical procedure behind the calibration of Section II and four checks of it: leave-one-out targeting, the shared transition of the rate and volatility schedules, the convenience-yield correction, and the sign of ρ_{cx}. It ends with the behavior of the targeted moments jointly across the model's samples (Section III.G).

Table CI: Return–volatility correlations by horizon. "Data" is the correlation between NY Fed primary-dealer equity returns and changes in their one-year rolling volatility, computed over non-overlapping windows of the stated horizon (2000–2018, 216 monthly observations). The model columns apply the identical estimator to wealth returns simulated from the calibrated model at two values of the underlying consumption–state correlation. Means across 400 simulated samples, standard deviations in parentheses.

Horizon	Data	Baseline model, $\rho_{cx} = -1$	Model, $\rho_{cx} = -0.3$
1 month	-0.22	-0.08 (0.08)	-0.05 (0.08)
3 months	-0.37	-0.18 (0.12)	-0.12 (0.13)
6 months	-0.48	-0.29 (0.15)	-0.18 (0.18)
12 months	-0.73	-0.51 (0.20)	-0.33 (0.23)

D.1 Numerical procedure

After the directly set and the measured parameters are fixed, the two free rates $\theta = (r_{\min}, r_{\max})$ minimize the largest standardized deviation of the eight targeted moments from the data,

$$\hat{\theta} \;=\; \arg\min_{\theta} \; \max_{k} \; \frac{\left| m_k(\theta) - \hat{m}_k \right|}{\hat{\sigma}_k}, \tag{D1}$$

where $m_k(\theta)$ is the model's value of moment k, $\hat{m}_k$ its value in the data, and $\hat{\sigma}_k$ its model-implied standard error. During the search, each $m_k(\theta)$ is the mean across 200 simulated 35-year samples (the TIPS moments over the last 21 years of each). The samples are drawn with the same random numbers at every θ, so that differences in the criterion across parameter values are not driven by simulation noise. The standard errors are held fixed during the search, at the standard deviations of the moments across 2,000 samples at a preliminary calibration, which lie between 0.92 and 1.00 times those at the final estimate. Table II reports the moments and standard errors evaluated again on 2,000 samples at the chosen rates.

The search starts from the nine points of a 3×3 grid over the two rates, with $r_{\min}$ at -0.6%, 0.0% and 0.6% and $r_{\max}$ at 6.6%, 7.5% and 9.0%. From each, the covariance matrix adaptation evolution strategy (CMA-ES) searches the whole box, and a local refinement by sequential least-squares quadratic programming (SLSQP, in NLopt, Johnson 2007) finishes it. Asset prices at a parameter set are well defined when the dividend strips decay quickly enough for the P/D ratio to be finite, at a long-run rate of at least 0.1% a year. All nine runs

reach the same optimum, $r_{\min} = -0.19\%$ and $r_{\max} = 9.15\%$, where the largest deviation, 1.46 standard errors, is shared by the equity premium, the mean P/D ratio and the volatility of returns. This minimization is the inner step of the joint measurement and calibration fixed point of Appendix C.

The criterion is evaluated over a wide box of the two rates, with $r_{\min}$ from -3% to 4% and $r_{\max}$ from -3% to 30%, at the points of a 50×50 grid, of which 1,964 give well-defined asset prices, and Figure 2 shows the region around the valley. The criterion has a single local minimum, and a finer grid around the estimate finds no lower value. The rates at which the criterion is below 1.96, so that all eight moments lie within 1.96 frozen standard errors, form a band around the estimate, with $r_{\min}$ between about -0.6% and 1.0% and $r_{\max}$ between about 7.3% and 11.5%.

The largest deviation also gives a joint test of the eight moments, which accounts for their being judged together. In units of the standard deviations across the model's 2,000 samples at the estimate, as in Table II, the largest of the eight deviations of the data from the model is 1.42, for the mean P/D ratio. Each simulated sample also has a largest deviation from the model's means, and in 56% of the samples it exceeds 1.42, so by this measure the data look like a typical sample from the model. Across the samples, the 95th percentile of the largest deviation is 2.87, against 1.96 for a single moment.

D.2 Leave-one-out

Table DI chooses the two rates again eight times, each time dropping one of the targeted moments and keeping the same standard errors and simulated samples, and reports all eight moments in each run. In five of the eight cases the rates do not move, because the dropped moment does not bind at the estimate: the mean risk-free rate, the volatility of the P/D ratio, the two TIPS moments and the predictability slope. In all eight cases, every remaining moment stays inside its 95% interval, and in seven the dropped moment also lands inside its own. With the equity premium dropped, the rates move to 0.23% and 8.83%, and the model still delivers a premium of 4.8%, close to the 5.3% of the baseline model.

The exception is the mean P/D ratio. Without it, the rates fall to -0.70% and 5.70%, and the other seven moments fit about as well as in the baseline model, with a largest deviation of 1.48 against 1.44, evaluated on 2,000 samples in units of the standard errors frozen for

the search. The mean P/D ratio, however, rises to 195.0, against 77.7 in the baseline model and 53.8 in the data (Table DI). The other seven moments therefore leave the level of the rate schedule largely open, and the mean P/D ratio pins it down: lower real rates lower the discount rate on dividends and raise valuations.

Table DI: Leave-one-out. Each of the last eight rows drops the targeted moment that labels it, shown in brackets, and chooses the two rates again against the remaining seven, with the same standard errors and simulated samples. The moments and their symbols are those of Table II, and the model's values are means across 2,000 samples. An asterisk marks a value outside the model's 95% interval in that run. The SE row gives each moment's model-implied standard error at the baseline, as in Table II. The SEs of each run, which set its asterisks, are within 16% of these, except in the run without the mean P/D ratio, where the SE of the mean P/D ratio is 2.5 times as large. The rates are in % p.a. Throughout, $\rho_{cx} = -1$ and $\rho = 0.01$.

	$\bar{r}_e$	$\overline{P/D}$	$\bar{r}_f$	σ_e	med σ_{pd}^{2y}	TIPS sp.	TIPS rx.	$\hat{\beta}_1$	$r_{\min}$	$r_{\max}$
Data	8.09	53.8	0.67	15.47	0.067	0.46	0.71	0.215		
Baseline	5.30	77.7	1.15	19.95	0.093	0.51	0.92	0.237	−0.19	9.15
SE	2.19	16.7	0.92	3.38	0.021	0.06	0.52	0.128		
$\bar{r}_e$	[4.82]	70.5	1.47	18.57	0.087	0.47	0.87	0.236	0.23	8.83
$\overline{P/D}$	5.31	[195.0*]	0.23	19.93	0.093	0.37	0.69	0.229	−0.70	5.70
$\bar{r}_f$	5.30	77.7	[1.15]	19.95	0.093	0.51	0.92	0.237	−0.19	9.15
σ_e	5.40	76.9	1.14	[20.25]	0.094	0.52	0.95	0.238	−0.25	9.40
med σ_{pd}^{2y}	5.30	77.7	1.15	19.95	[0.093]	0.51	0.92	0.237	−0.19	9.15
TIPS sp.	5.30	77.7	1.15	19.95	0.093	[0.51]	0.92	0.237	−0.19	9.15
TIPS rx.	5.30	77.7	1.15	19.95	0.093	0.51	[0.92]	0.237	−0.19	9.15
$\hat{\beta}_1$	5.30	77.7	1.15	19.95	0.093	0.51	0.92	[0.237]	−0.19	9.15

D.3 The shared-transition restriction

In the baseline, the rate schedule $r(x)$ in (B3) and the volatility schedule $\sigma_c(x)$ in (B2) share the same center x^* and width w, so both transitions occur over the same band of states. To check this restriction, a parameter k scales the width of the rate schedule, $r(x) = r_{\min} + (r_{\max} - r_{\min})\, L((x - x^*)/(kw))$, so that $k = 1$ is the baseline, and the two rates are chosen again at other values of k. The fit is worse for wider transitions. The largest deviation in the criterion of the search rises from 1.46 at $k = 1$ to 1.48 at $k = 1.5$ and 1.75

at $k = 3$. The fit is marginally better for narrower ones, at 1.42 for $k = 0.5$. The shared transition, $k = 1$, is kept as the simpler and more parsimonious specification.

D.4 The convenience-yield correction

As described in Section II, the model's $r(x)$ is the riskless rate the intermediary itself faces, while the empirical target is the one-month Treasury bill rate less inflation. Bills carry a convenience yield that a leveraged intermediary does not earn, so the baseline raises the target for the mean real rate by the 65 basis points estimated by Van Binsbergen *et al.* (2022) and lowers the target for the equity excess return, measured against that rate, by the same amount. Because the correction moves only these two targets, the calibration can simply be repeated without it.

Table DII compares the two. Without the correction, the targets are 0.02% for the mean real rate and 8.74% for the premium, and the rates move only slightly, to −0.37% and 9.32%. The model's mean real rate falls from 1.15% to 1.03%, about one standard error above the uncorrected target, and its premium rises from 5.30% to 5.52%. All eight moments remain inside their 95% intervals, so the calibration does not rest on the correction.

D.5 The sign of the consumption–state correlation

The link between consumption and wealth in Section I.F fixes the magnitude $|\rho_{cx}| = 1$, since a single innovation drives both wealth and the state, but the negative sign is a modeling choice, motivated by the countercyclical leverage of intermediaries at the holding-company level (Section II). To check the alternative, the whole joint measurement and calibration is repeated with $\rho_{cx} = +1$. The volatility schedule is measured again under the positive-sign wealth mapping, and the two rates are chosen again against the same eight moments, with the rate schedule free to rise or fall with the state.

Table DIII reports the result. The positive sign requires a falling rate schedule, with the real rate declining from 5.3% in calm states to −8.5% in the most volatile ones. Even then it matches only five of the eight moments. The volatility of returns falls to 9.9%, against 15.5% in the data, the volatility of the P/D ratio to 0.047, against 0.067, and the mean P/D ratio rises to 90.0. No pair of rates in the search box brings all eight moments inside their 95% intervals. The dealer data also favor the negative sign. Dealer returns are negatively

Table DII: Sensitivity to the convenience-yield correction. The 65 bp column is the baseline. The 0 bp column chooses the two rates again with the uncorrected targets for the mean real rate and the equity premium, holding everything else fixed. Rates and returns in %.

	Convenience-yield correction	
	0 bp	65 bp
Targets		
Mean real rate (%)	0.02	0.67
Equity premium (%)	8.74	8.09
Estimated rates		
$r_{\min}$ (%)	−0.37	−0.19
$r_{\max}$ (%)	9.32	9.15
Model moments		
Mean real risk-free rate (%)	1.03	1.15
Equity excess return (%)	5.52	5.30
Mean P/D ratio	80.9	77.7
Std. stock return (%)	20.58	19.95
Median 2-yr std $\log(P/D)$	0.096	0.093
TIPS 10–5 yr yield spread (%)	0.52	0.51
TIPS 10–5 yr excess return (%)	0.95	0.92
Return pred. slope	0.238	0.237

correlated with changes in their volatility at every horizon, from −0.22 at one month to −0.73 at twelve (Table CI). The baseline model reproduces the sign at every horizon, with smaller magnitudes, while the positive sign reverses it, to between +0.10 and +0.56. The sign follows largely from the assumption itself, so this is evidence for the choice of ρ_{cx} rather than an independent test of the model.

D.6 The Sharpe ratio and valuations across samples

Figure D1 shows the Sharpe ratio of the baseline model against the data, and how far the two rates can raise the equity premium (Section III.G). The data lie at the edge of the cloud of simulated samples. Only 1.9% of the samples lie above the line of the data's Sharpe ratio, so the observed Sharpe ratio falls just outside the baseline model's 95% interval. The pair of

moments, judged jointly, lies inside its 95% region.

The figure also shows how far the two rates can raise the premium. Each green dot is the model at one pair of rates $(r_{\min}, r_{\max})$, its mean premium and return volatility over the same 2,000 samples. The pairs are those of the grid in Appendix D.1, which covers the whole range of rates the estimation searches, and a pair is shown when the other seven targeted moments lie inside their 95% intervals by their own standard errors, as in Table II. These 1,382 pairs lie on nearly one curve. Whatever the rates, the premium and the volatility of returns move together, from a volatility of 15.7% to 24.1%, and the Sharpe ratio is at most 0.28, well below that of the data. The largest of these premia, 6.8% (green diamond, located more finely than the grid, at $r_{\min} = -1.30\%$ and $r_{\max} = 10.16\%$), is reached where the mean P/D ratio and the 10–5 yr TIPS spread both reach the upper ends of their intervals. Further along the same curve, where these moments leave their intervals, the premium reaches its observed value of 8.1% only at a return volatility of 28%.

Figures D2.A and D2.B compare the data with the model's samples for every pair of the eight targeted moments (Section III.G), the first for the 13 pairs that include the mean P/D ratio or the equity premium, and the second for the other 15. For each pair, the model's 95% region is the ellipse that holds 95% of the samples, in the metric of the two moments' covariance across samples. The data lie inside it in 25 of the 28 pairs, including all 15 in Figure D2.B. The three exceptions all involve the mean P/D ratio, paired with the risk-free rate, the volatility of returns, and the volatility of the P/D ratio. Across the samples, the mean P/D ratio is strongly negatively correlated with each of these. At the P/D ratio of the data, the samples have a risk-free rate of about 2.4% and a return volatility of about 24%, against 0.7% and 15.5% in the data.

Table DIII: The sign of ρ_{cx}. The $\rho_{cx} = +1$ column repeats the full measurement and calibration with the sign flipped and the rate schedule free to fall. Standard errors across the model's 2,000 samples at each calibration (400 for the return–volatility correlations) in parentheses. The Matched row counts the moments inside their 95% intervals by these standard errors. The largest deviations are evaluated on the 2,000 samples, in units of the standard errors frozen for the search (Appendix D.1), so the baseline's differ slightly from the 1.46 of the search itself, which uses 200 samples. These are the same in both columns, while the standard errors in parentheses are each calibration's own. Far from the baseline the two can differ widely: in the $\rho_{cx} = +1$ column, the volatility of returns lies within two frozen standard errors of its target but more than ten of its own. There, $r_{\min}$ and $r_{\max}$ remain the rates at low and high volatility, so the schedule falls. Rates and returns in %.

Moment	Data	$\rho_{cx} = -1$	$\rho_{cx} = +1$
Equity excess return (%)	8.09	5.30 (2.19)	3.10 (2.70)
Mean P/D ratio	53.8	77.7 (16.7)	90.0 (6.1)
Median 2-yr std $\log(P/D)$	0.067	0.093 (0.021)	0.047 (0.005)
Std. stock return (%)	15.47	19.95 (3.38)	9.91 (0.47)
Mean real risk-free rate (%)	0.67	1.15 (0.92)	1.42 (2.24)
TIPS 10–5 yr yield spread (%)	0.46	0.51 (0.06)	0.55 (0.47)
TIPS 10–5 yr excess return (%)	0.71	0.92 (0.52)	0.41 (0.31)
Return pred. slope	0.215	0.237 (0.128)	0.164 (0.164)
Return–volatility correlation (not targeted)			
1 month	−0.22	−0.08 (0.08)	+0.10 (0.08)
3 months	−0.37	−0.18 (0.12)	+0.23 (0.13)
6 months	−0.48	−0.29 (0.15)	+0.36 (0.16)
12 months	−0.73	−0.51 (0.20)	+0.56 (0.19)
Matched (95% interval)	—	8 of 8	5 of 8
Largest deviation (SEs)	—	1.53	2.36
Second-largest deviation (SEs)	—	1.44	2.32
$r_{\min}$ (%)	—	−0.19	5.35
$r_{\max}$ (%)	—	9.15	−8.51
Rate schedule	—	increasing	decreasing

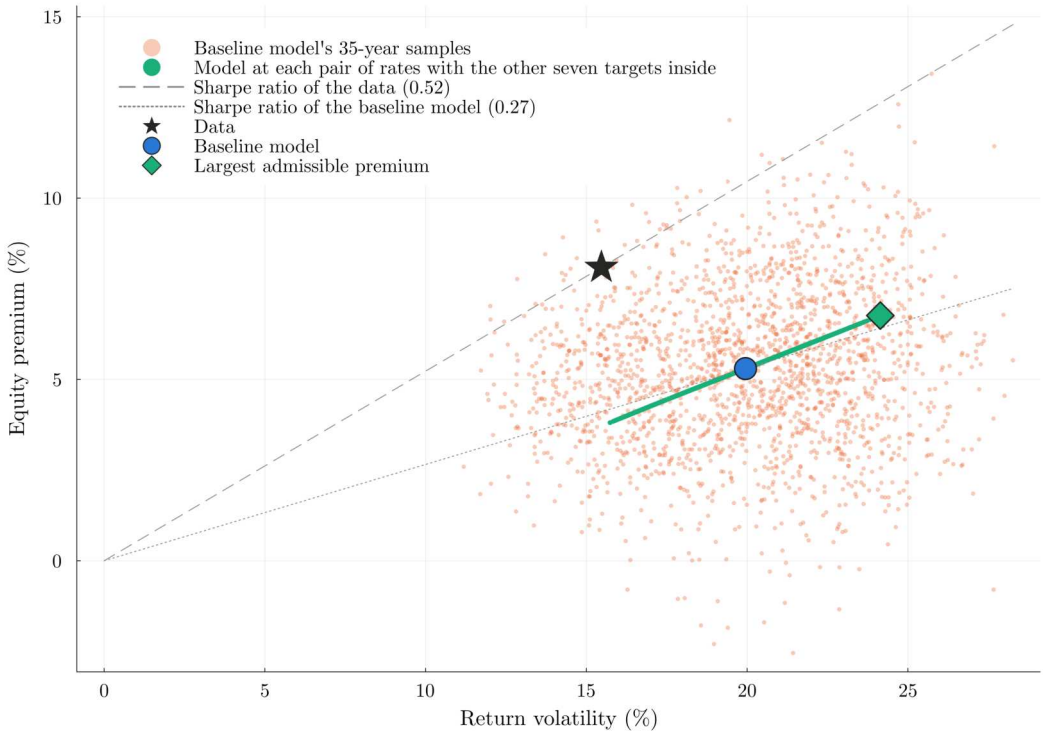


Figure D1: The Sharpe ratio of the baseline model against the data, with the equity premium against return volatility. The light orange points are the baseline model's 2,000 simulated 35-year samples, and the blue circle is their mean, the value in Table II. Each green dot is the model's mean at one pair of rates $(r_{\min}, r_{\max})$ of the grid in Appendix D.1, shown when the other seven targeted moments lie inside their 95% intervals, and the green diamond is the largest such premium. All points use the same 2,000 simulated paths. The two lines through the origin are the Sharpe ratios of the data and of the baseline model, and only 1.9% of its samples lie above the line of the data.

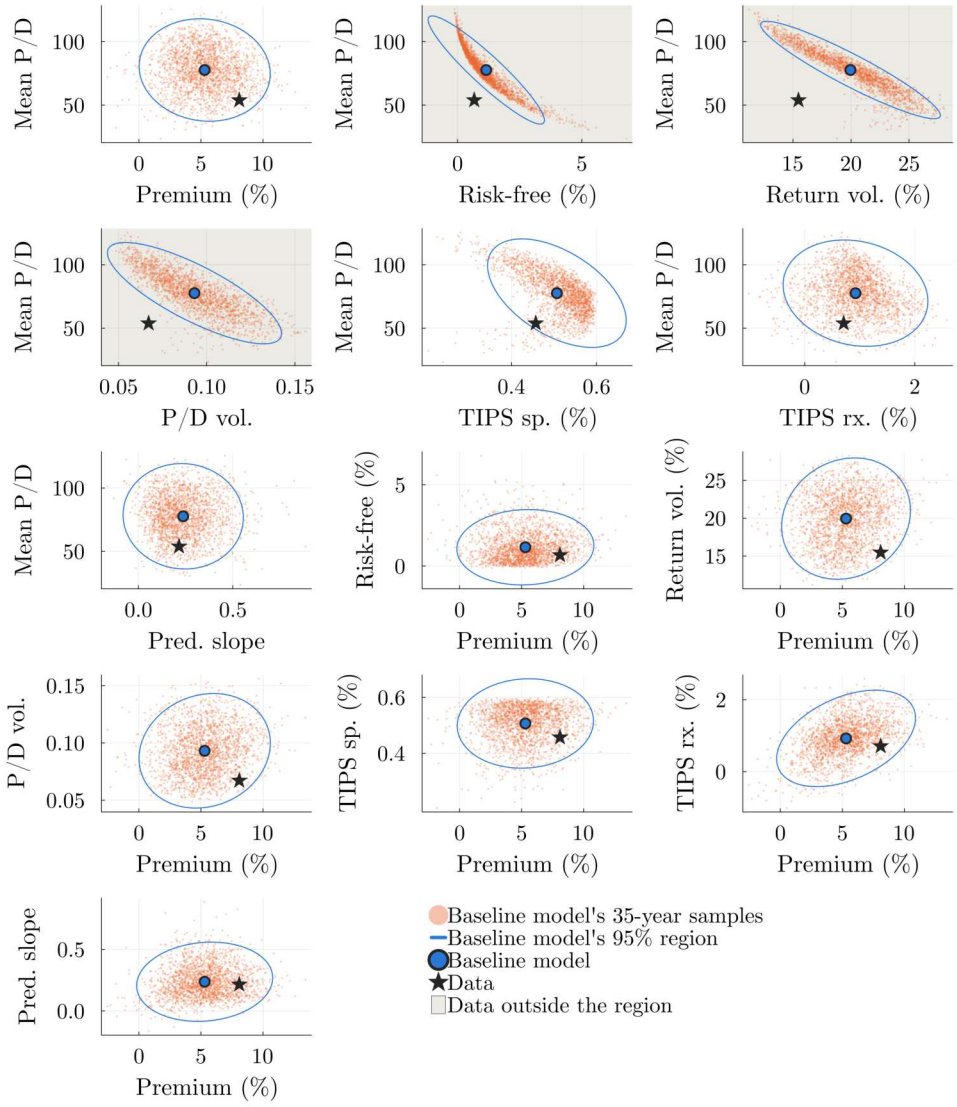


Figure D2.A: The data against the model's samples, for the pairs of targeted moments that include the mean P/D ratio or the equity premium. Each light orange point is one of the baseline model's 2,000 simulated 35-year samples (the TIPS moments over the last 21 years of each), the blue circle is their mean, the value in Table II, and the star is the data. The blue ellipse is the model's 95% region, holding 95% of the samples in the metric of the pair's covariance across samples. Shaded panels are those where the data lie outside it.

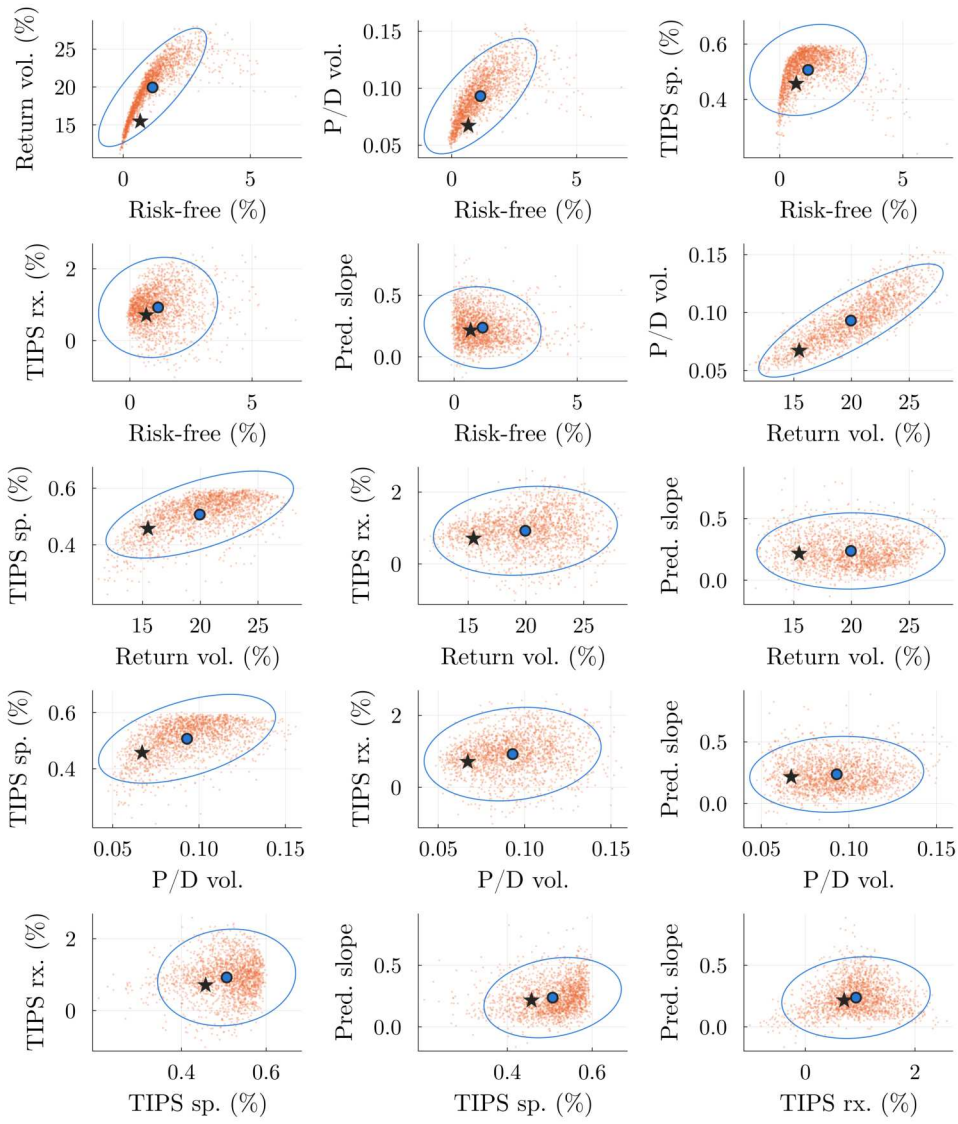


Figure D2.B: The data against the model's samples, for the pairs of targeted moments that include neither the mean P/D ratio nor the equity premium. Each light orange point is one of the baseline model's 2,000 simulated 35-year samples (the TIPS moments over the last 21 years of each), the blue circle is their mean, the star is the data, and the blue ellipse is the model's 95% region, as in Figure D2.A.

Appendix E. Asset-Price Evidence from the Recovered State

The state variable is measured from the volatility of dealer equity returns to test the conditional predictions of the model. The dealer returns come from the price of a traded equity portfolio, so the measurement is not entirely free of price information. It uses only the volatility of those returns, however. No price level, valuation ratio, or yield enters the measurement, including the series it is compared with below. At each date the implied state x_t is recovered by inverting the model-implied wealth volatility $\sigma_N(x)$ at the observed dealer volatility. Bond and stock prices are then evaluated at that state. Figure E1 shows the dealer volatility series, the state recovered from it, and the implied price–dividend ratio, 10-year TIPS yield, and 10–5 yr TIPS spread against their empirical counterparts. Table EI reports the corresponding correlations.

The model-implied prices comove positively with observed equity valuations and the slope of the real term structure (Section III.F). The implied price–dividend ratio has a levels correlation of 0.30 with the observed ratio, and the implied TIPS spread one of 0.50. The 10-year TIPS yield is the one series the mapping does not track, with correlations of 0.04 in levels and 0.03 in changes. Several broad movements are nonetheless reproduced. The model-implied price–dividend ratio falls sharply into the trough of early 2009 and recovers alongside its empirical counterpart, though its swings are considerably larger than those in the data. The implied TIPS spread, by contrast, stays close to the average level of the observed spread but is much smoother.

Table EI: Dealer volatility and model-implied asset prices. Panel A reports summary statistics for the 1-year rolling equity return volatility of NY Fed primary dealer holding companies, computed from the He *et al.* (2017) daily return series. Panel B reports the correlations of the model-implied series with their empirical counterparts, in levels and in monthly changes. Panel A and the correlations with the P/D ratio cover December 2000 to November 2018, and the correlations with TIPS yields January 2004 to November 2018, the TIPS sample of the calibration targets.

Panel A: Intermediary equity return vol (% p.a.)

Mean	Std	Min	Max	AR(1)[a]
23.1	13.7	8.3	72.6	0.891

Panel B: Correlations, model-implied vs. empirical

Series	Levels	Changes
P/D ratio	0.305	0.386
10-yr TIPS yield	0.040	0.028
10–5yr TIPS spread	0.502	0.148

[a]AR(1) on monthly series. Reported value is a^{12}, the implied annual persistence, where a is the monthly AR(1) coefficient.

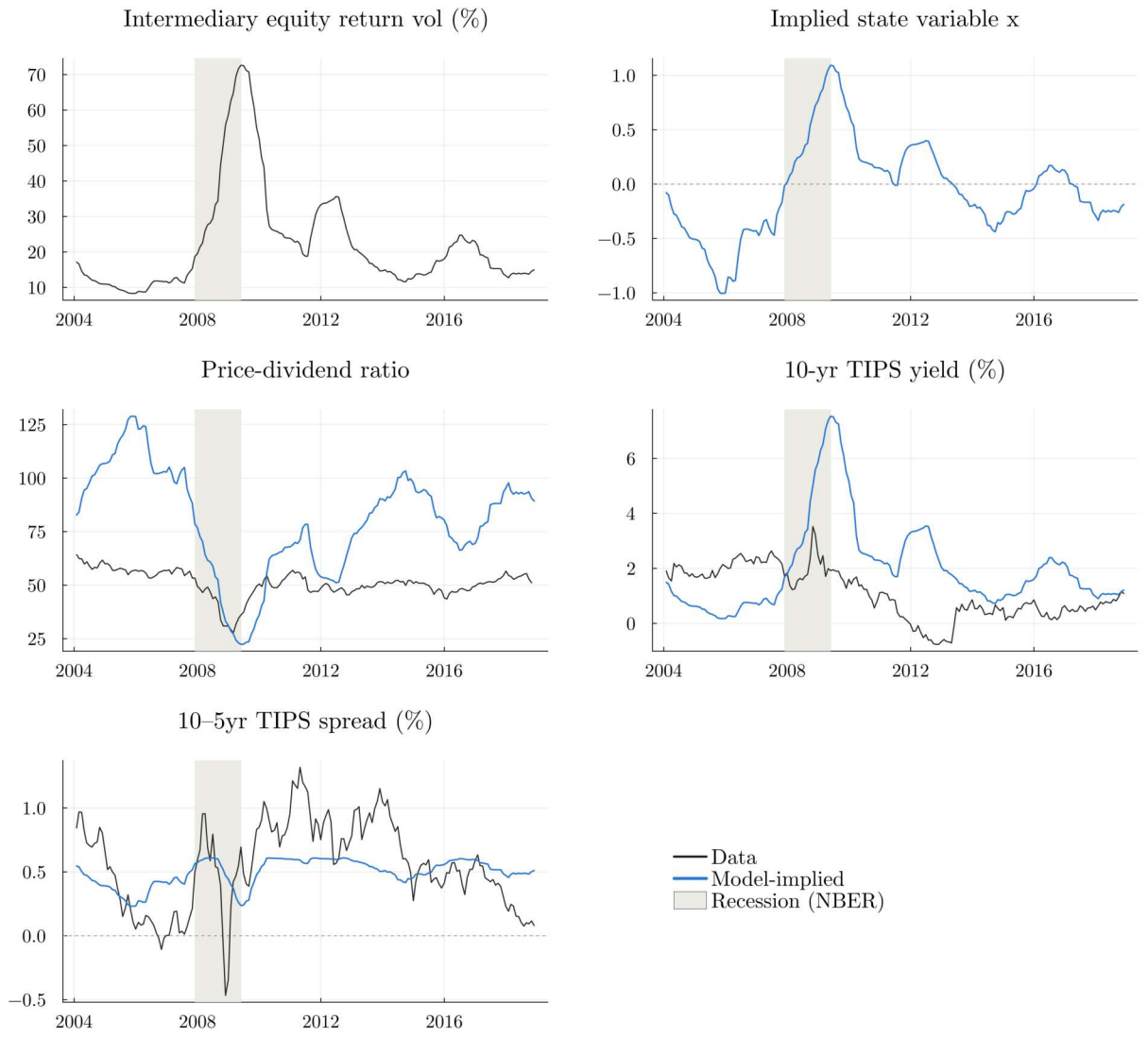


Figure E1: Intermediary-implied state variable and asset prices. Top left: 1-year rolling equity return volatility of NY Fed primary dealer holding companies from He *et al.* (2017). Top right: the implied state variable x_t of the model, obtained by inverting the model-implied wealth volatility $\sigma_N(x)$ at each date using the empirical equity return volatility. Middle left/right and bottom: model-implied P/D ratio, 10-year TIPS yield, and 10–5 yr TIPS spread (blue) overlaid on their empirical counterparts (black). Shaded bands mark NBER recessions.

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Internet Appendix

High Volatility Is the Key: A Joint Treatment of Asset Pricing Puzzles

IA.A Simulation

Figure IA.1 presents a simulated forty-year path of the baseline model's consumption dynamics and the asset prices implied by them.

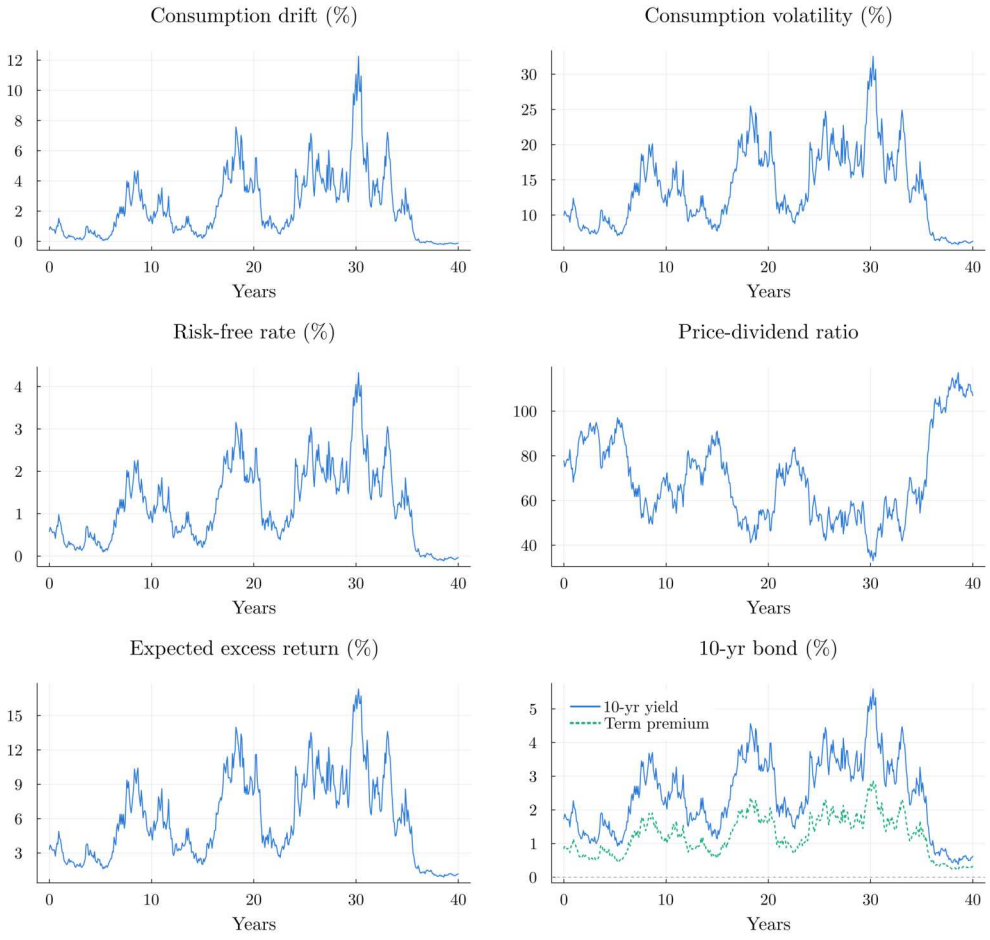


Figure IA.1: Variables based on a simulated path for forty years under the baseline parameterization. The panels show the consumption drift μ_c, consumption volatility σ_c, the risk-free rate r, the price–dividend ratio, the expected stock excess return, and the 10-year real bond yield together with its term premium.

IA.B Expectations-Hypothesis Regression Results

Tables IA.1 and IA.2 report the FB and CS slopes of the model, with their dispersion across samples, against the published estimates. Section III.E discusses them.

Table IA.1: Fama–Bliss expectations-hypothesis regressions. The one-year log excess return of an n-year zero-coupon bond regressed on the spread between its forward rate and the one-year yield, $f_t(n) - y_t(1)$, with monthly overlapping observations. Model: mean slope across 2,000 simulated 35-year samples, with its standard deviation across samples in parentheses. Data: the nominal Treasury estimates of Cochrane and Piazzesi (2005, their Table 2) for 1964 to 2003, as reported by Gabaix (2012), with small-sample standard errors in parentheses. The expectations hypothesis implies a slope of zero.

Maturity	Model	Data
2-year	2.54 (1.99)	0.99 (0.33)
3-year	2.58 (2.41)	1.35 (0.41)
4-year	2.60 (2.91)	1.61 (0.48)
5-year	2.59 (3.46)	1.27 (0.64)

Table IA.2: Campbell–Shiller expectations-hypothesis regressions. The one-month change in the yield of an n-month bond, scaled by its remaining maturity, $(n-1)\,[y_{t+1}(n-1) - y_t(n)]$, regressed on the spread $y_t(n) - y_t(1)$. Model as in Table IA.1. Data: the nominal Treasury estimates of Campbell *et al.* (1997, their Table 10.3), as reported by Gabaix (2012), with standard errors in parentheses. The expectations hypothesis implies a slope of one.

Maturity	Model	Data
3-month	−4.35 (3.37)	−0.15 (0.28)
6-month	−4.38 (3.45)	−0.83 (0.44)
1-year	−4.44 (3.62)	−1.43 (0.60)
2-year	−4.56 (4.02)	−1.45 (1.00)
4-year	−4.77 (5.03)	−2.27 (1.46)

IA.C Recursive Preferences

The baseline model uses CRRA utility. This appendix explains why recursive utility, the framework of the long-run risk models (Bansal and Yaron 2004), does not by itself generate positive real term premia, despite its success in explaining the equity premium. Long-run risk models feature a time-varying consumption drift, and possibly a time-varying consumption diffusion. Since they use a representative agent, the drift and diffusion take values close to those observed in aggregate data.

Following Duffie and Epstein (1992), when utility is recursive, lifetime utility is:

$$V_0 = \mathrm{E}_0 \int_0^\infty F(C_t, V_t)\mathrm{d}t \tag{IA.1}$$

where V_t is the value function at time t, and $F(\cdot,\cdot)$ determines flow utility. The latter is also the aggregator of the recursive utility process, and it is given by:

$$F(C_t, V_t) = \frac{\rho\left(1-\gamma\right)V_t}{1-1/\psi}\left(\left(\frac{C_t}{\left(\left(1-\gamma\right)V_t\right)^{1/(1-\gamma)}}\right)^{1-1/\psi} - 1\right) \tag{IA.2}$$

where ψ is the intertemporal elasticity of substitution. The case of CRRA utility is nested in this specification, and it corresponds to the case where $\psi = 1/\gamma$. The SDF is derived from the expressions for the value function and the aggregator function. As shown by Tsai and Wachter (2018), and similarly elsewhere in the literature (Benzoni, Collin-Dufresne and Goldstein 2011; Chen, Cosimano, Himonas and Kelly 2014), the value function can be expressed as:

$$V_t = \frac{C_t^{1-\gamma} e^{(1-\gamma)K(x_t)}}{1-\gamma} \tag{IA.3}$$

where V_t increases with K, a function of x_t that captures the full dependence of the value function on the state variable. The model does not need to be solved fully for the analysis below. Given the expression for the value function, Itô's lemma can be applied to derive the SDF process. Following Chen *et al.* (2014), the fundamental relationship is:

$$\frac{\mathrm{d}\Lambda}{\Lambda} = F_V(C_t, V_t)\mathrm{d}t + \frac{\mathrm{d}F_C(C_t, V_t)}{F_C(C_t, V_t)} \tag{IA.4}$$

Here F_C and F_V denote partial derivatives of F with respect to consumption and the value function, respectively. Substituting Equation (IA.3) into both and applying Itô's lemma to F_C, with consumption and the state variable as independent variables, gives:

$$\begin{aligned}\frac{\mathrm{d}\Lambda}{\Lambda} =& \Bigg(\frac{\rho(-(1-\gamma\psi)e^{\frac{(1-\psi)K(x_t)}{\psi}} - \gamma\psi + \psi)}{1-\psi} - \gamma\mu_{ct} + \frac{\gamma^2\sigma_{ct}^2}{2} \\ &+ \frac{\gamma(\gamma\psi-1)\rho_{cx}\sigma_x\sigma_{ct}K'(x_t)}{\psi} \\ &+ \frac{(\gamma\psi-1)(-2\psi\log(\phi)x_tK'(x_t) + \sigma_x^2((\gamma\psi-1)K'(x_t)^2 - \psi K''(x_t)))}{2\psi^2} \Bigg)\mathrm{d}t \\ &- \frac{(\gamma\psi-1)\sigma_xK'(x_t)}{\psi}\mathrm{d}W_{xt} - \gamma\sigma_{ct}\mathrm{d}W_{ct} \end{aligned} \tag{IA.5}$$

When $\gamma = 1/\psi$ (time-separable utility), Equation (IA.5) reduces to Equation (6). For any value of ψ, the consumption risk term $-\gamma\sigma_{ct}\mathrm{d}W_{ct}$ is the same as in the time-separable case. Recursive utility adds the state-variable term $-\frac{(\gamma\psi-1)\sigma_xK'(x_t)}{\psi}\mathrm{d}W_{xt}$, which arises from the dependence of the value function on x_t through K.

Using the SDF expression, the risk-free rate, stock price, and bond price can be derived as before. The analysis focuses on the bond to show why recursive utility tends to generate a negative real term premium. Specifically, the expected excess return of the bond can be expressed as:

$$\mathrm{E}\left[\frac{\mathrm{d}B}{B}\right] - r\,\mathrm{d}t = \left(\rho_{cx}\gamma\sigma_{ct}\sigma_x + \frac{(\gamma\psi-1)\sigma_x^2K'(x_t)}{\psi}\right)\frac{B_x}{B}\,\mathrm{d}t \tag{IA.6}$$

The second term in parentheses is unique to recursive utility. Furthermore, its contribution to the expected excess return, once multiplied by B_x/B, is in general negative. Whether the state variable drives the consumption drift or the consumption diffusion, the following qualitative relationships typically hold in equilibrium:

$$\left.\begin{aligned}&(x \uparrow\Rightarrow \mu_c \uparrow) \Rightarrow (K'(x) > 0) \text{ and } (r'(x) > 0 \Rightarrow B_x < 0) \\ &(x \uparrow\Rightarrow \sigma_c \uparrow) \Rightarrow (K'(x) < 0) \text{ and } (r'(x) < 0 \Rightarrow B_x > 0)\end{aligned}\right\} \Rightarrow K'(x)B_x < 0 \tag{IA.7}$$

The first case corresponds to a growth shock. An increase in expected growth improves welfare ($K'(x) > 0$) but raises interest rates, causing bond prices to fall ($B_x < 0$). The second

case corresponds to a volatility shock. An increase in uncertainty reduces welfare ($K'(x) < 0$) but triggers precautionary savings, lowering rates and raising bond prices ($B_x > 0$). In both scenarios, the product $K'(x)B_x$ is negative. Given a preference for early resolution of uncertainty ($\gamma > 1/\psi \Rightarrow \gamma\psi - 1 > 0$), the recursive utility component in Equation (IA.6) therefore acts as a "hedging" term. This makes bonds an attractive asset for an investor worried about future states, thereby pushing the real term premium downward.[21]

Consequently, while recursive utility can be a powerful tool for generating a high equity premium, it tends to produce negative real term premia. A positive real term premium must then come from the consumption risk term, which is unchanged from the time-separable case and so needs the same conditions as there, namely high consumption volatility and a real rate that rises with it (Section I). This is consistent with Bansal and Shaliastovich (2013), whose long-run risk model generates a positive nominal term premium through an inflation channel while the real term premium remains negative. Ellison and Tischbirek (2021) document the same pattern across recursive-utility models with long-run risk and with rare disasters, tracing it to the same preference for early resolution of uncertainty. In Schneider (2026), too, the real term spread falls substantially when the elasticity of intertemporal substitution is raised above its time-separable value, with risk aversion held fixed. He attributes the fall to the elasticity itself, through a short rate that declines in bad states. The same change, however, introduces a preference for early resolution, and with it the hedging term above, which tends to lower the real term premium for given dynamics of the short rate.

[21] A fully general proof is not provided. The complication can arise from the fact that the risk-free rate is not only a function of consumption drift and diffusion, but also a function of the recursive utility term. In addition, the state variable can affect both consumption drift and consumption diffusion at the same time, and these relationships could in theory be arbitrarily non-linear. In these cases, it becomes more difficult to rigorously determine the sign of $K'(x)B_x$.